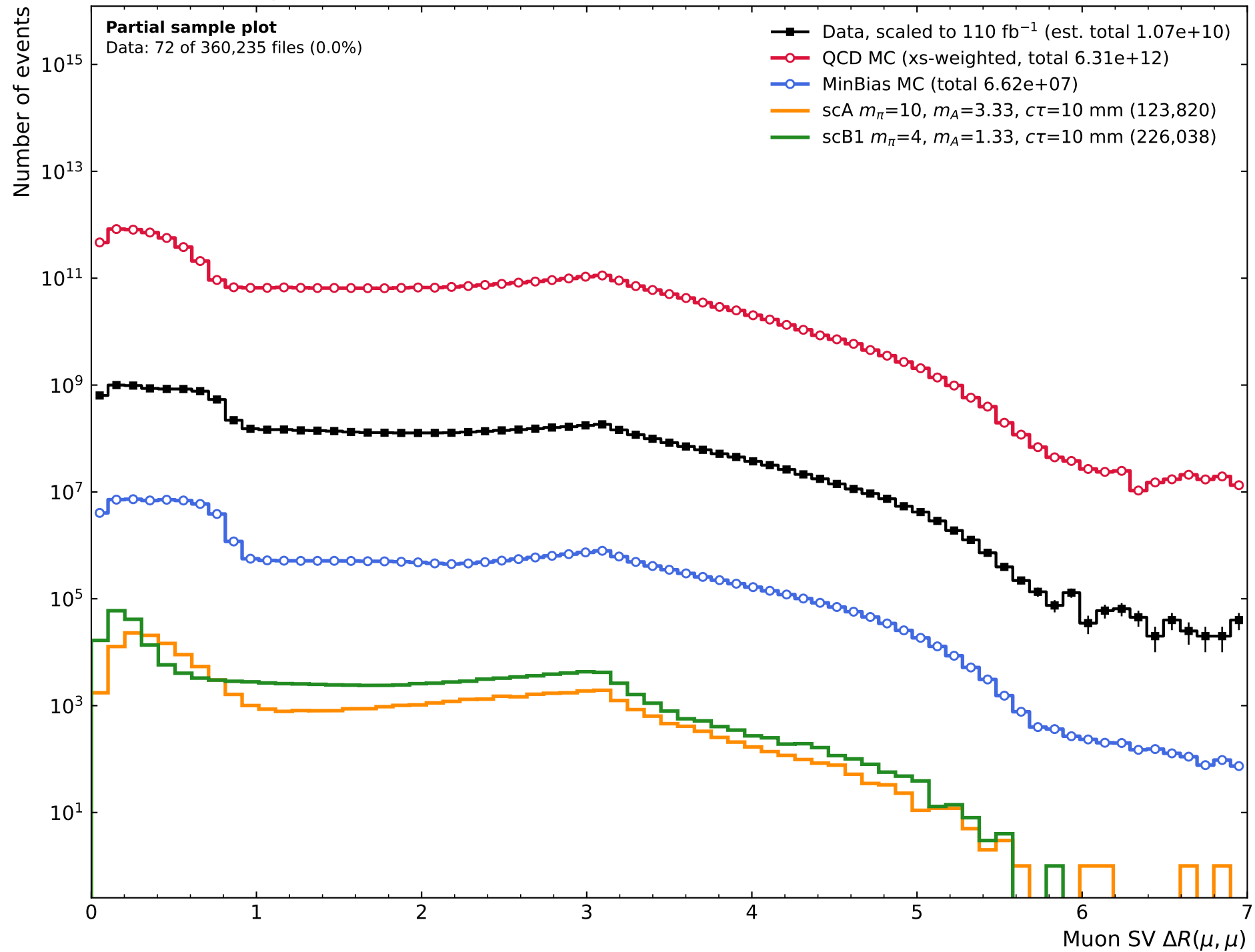


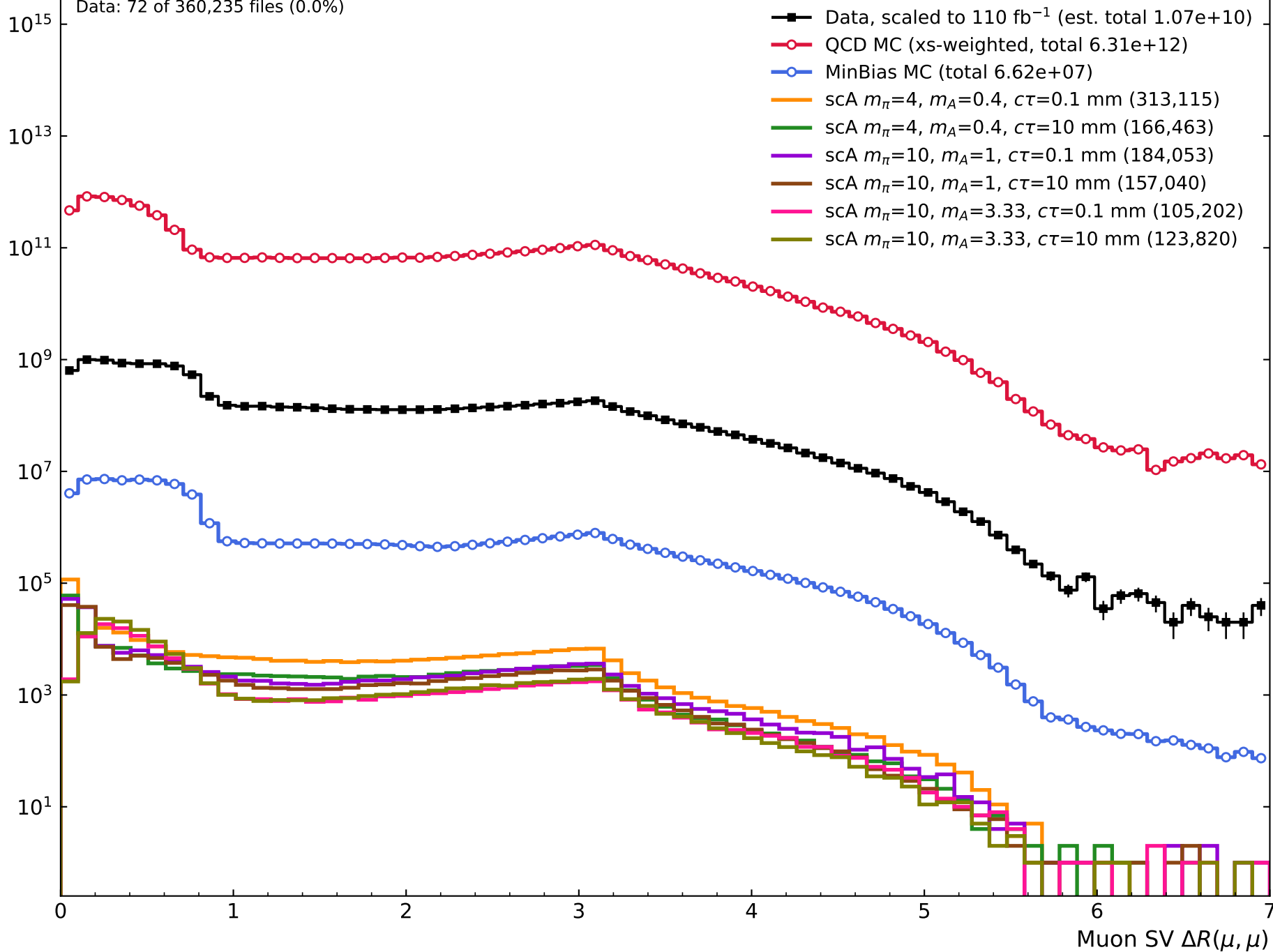


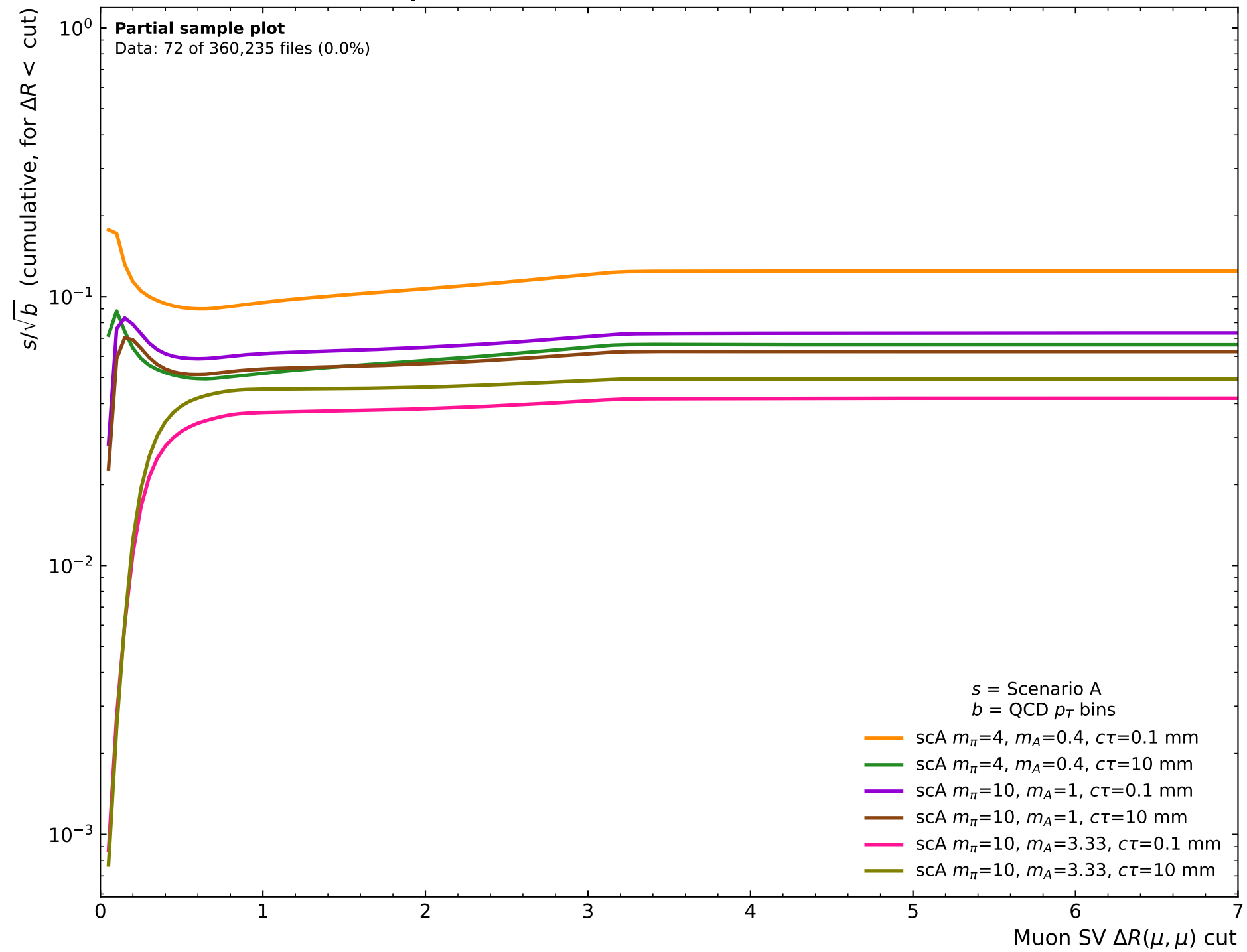
Number of events

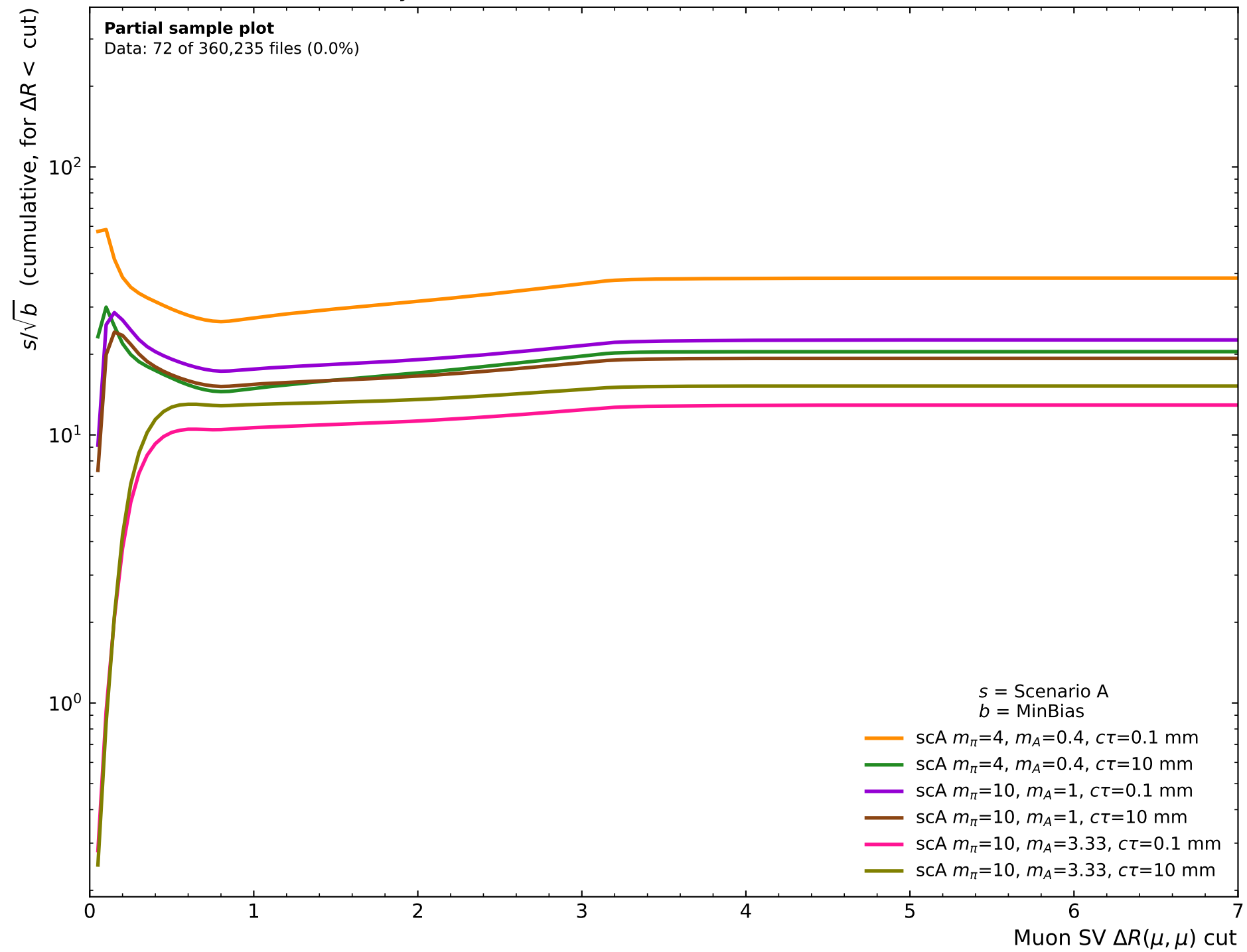
Partial sample plot

Data: 72 of 360,235 files (0.0%)

Scenario A





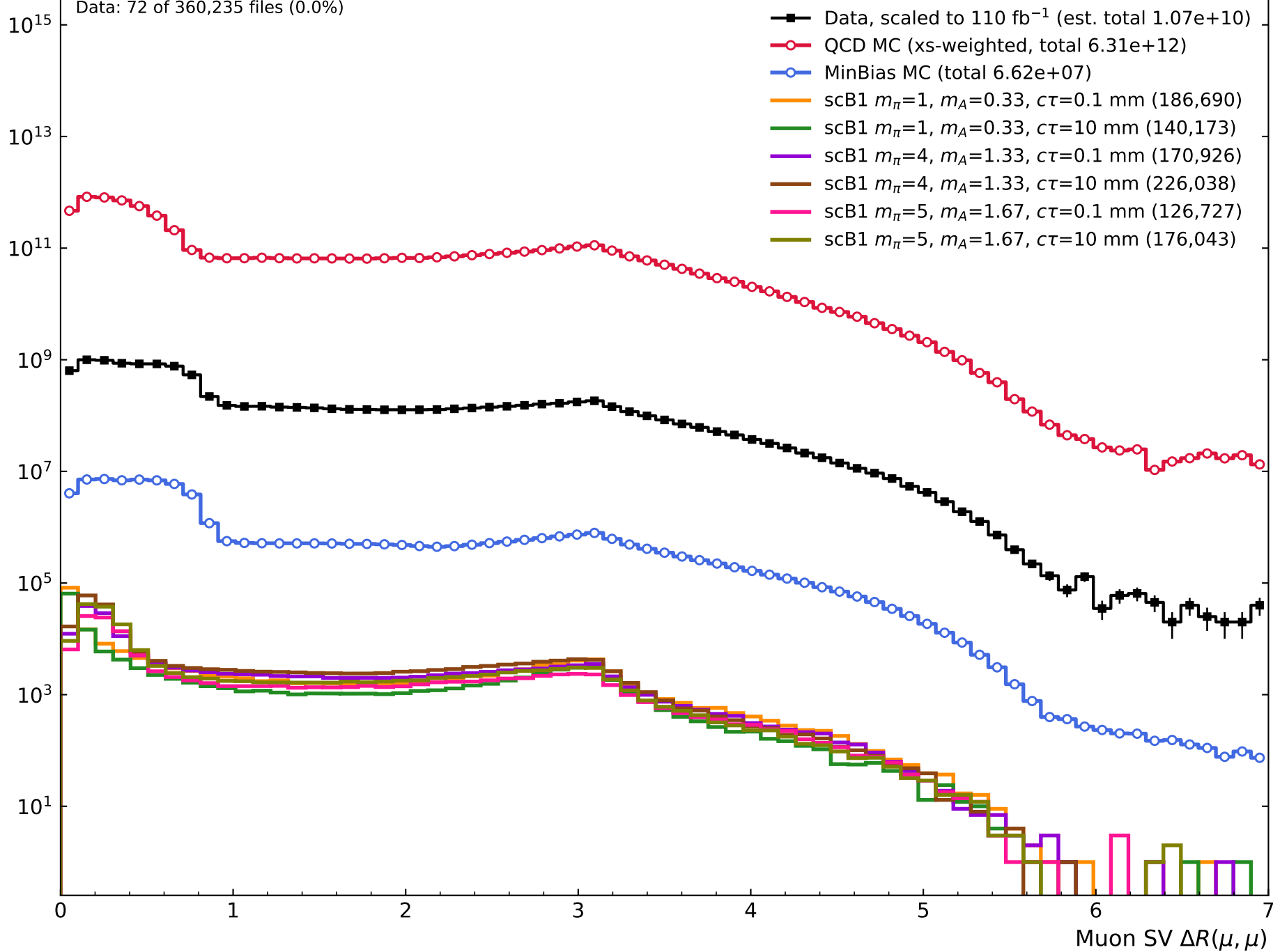


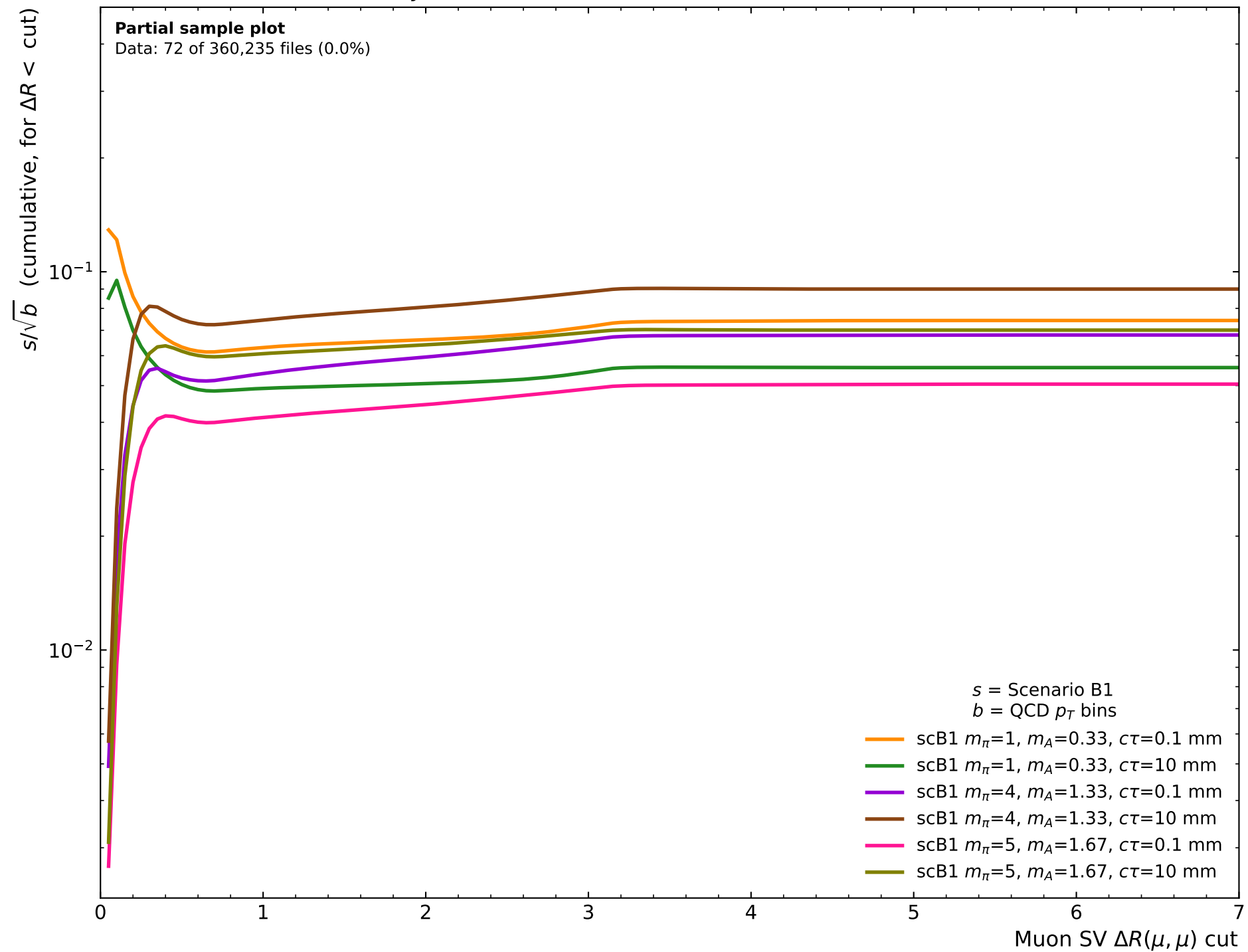
Number of events

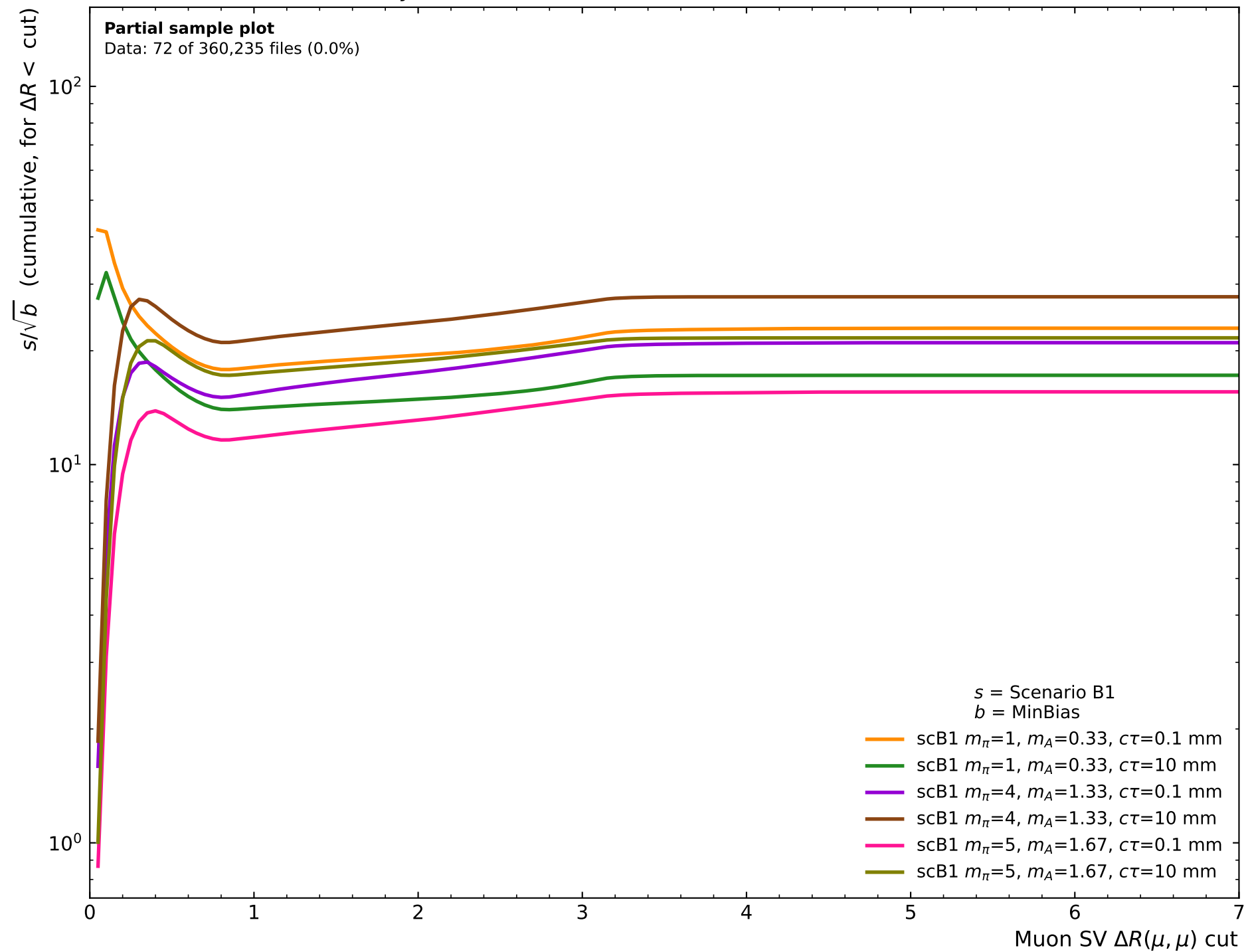
Partial sample plot

Data: 72 of 360,235 files (0.0%)

Scenario B1



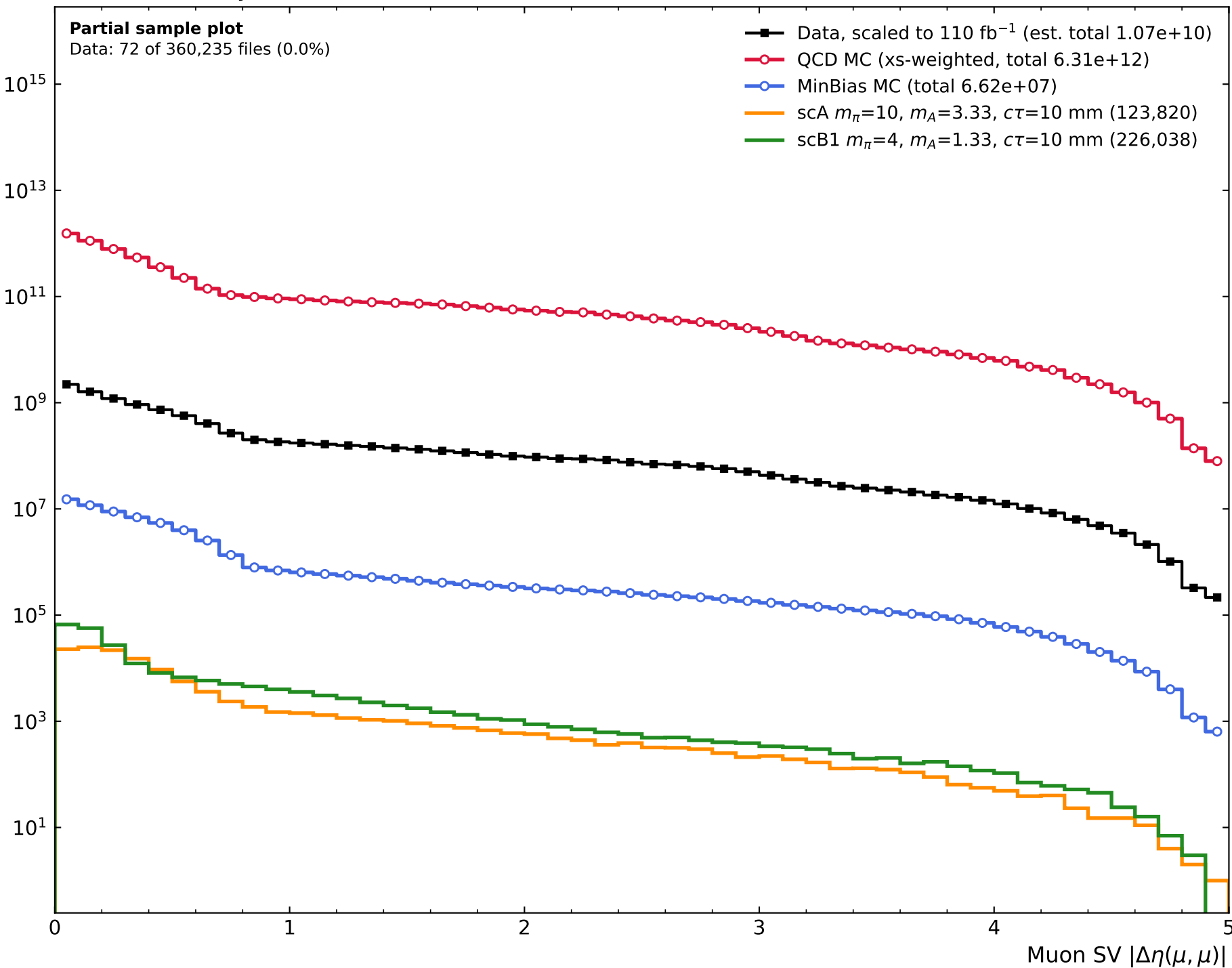




Number of events

Partial sample plot

Data: 72 of 360,235 files (0.0%)

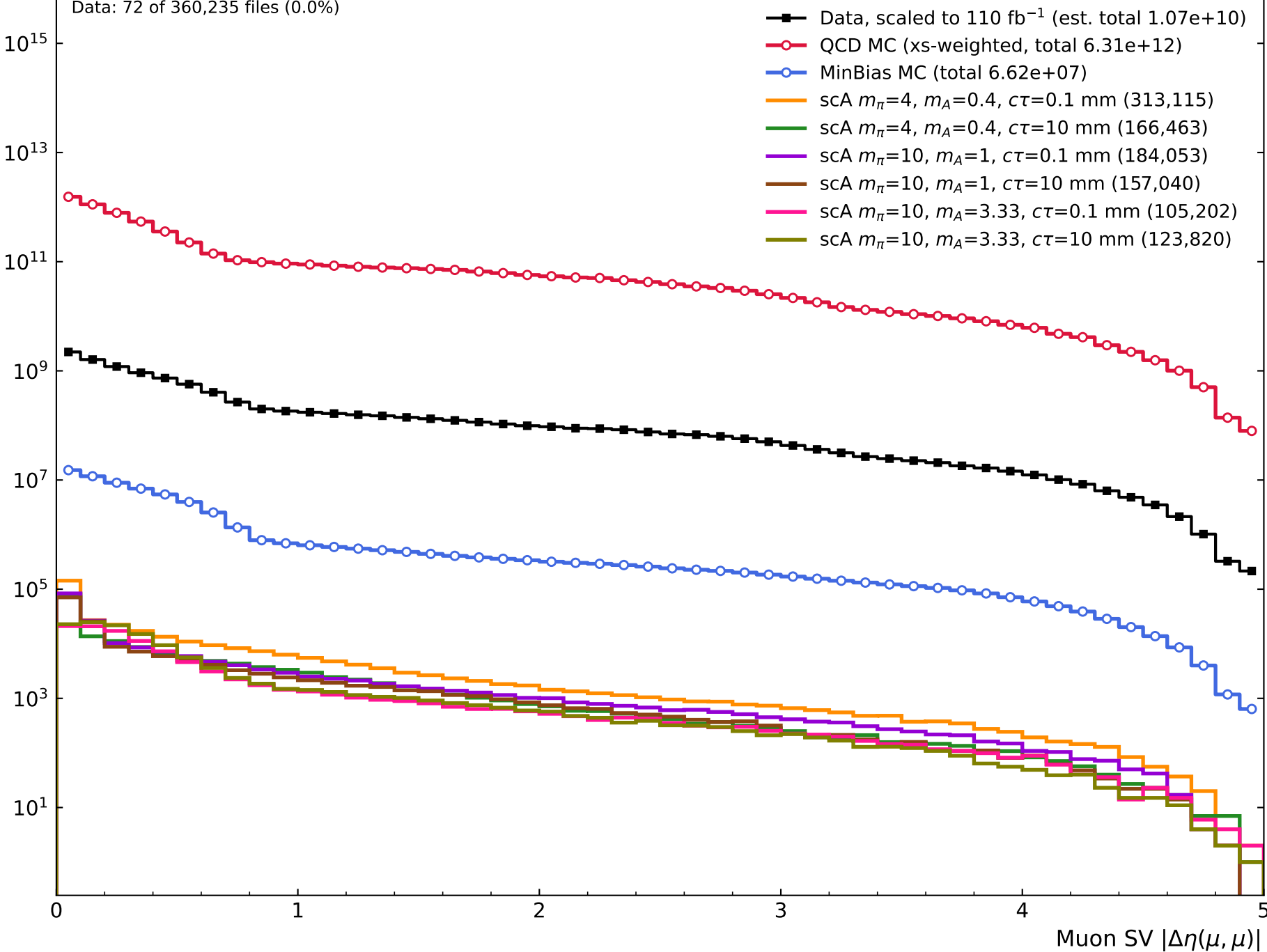


Number of events

Partial sample plot

Data: 72 of 360,235 files (0.0%)

Scenario A



$s/\sqrt{b}$ (cumulative, for $|\Delta\eta| < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)

Muons SV $|\Delta\eta(\mu, \mu)|$ cut

- s = Scenario A
 b = QCD p_T bins
- scA $m_\pi=4$, $m_A=0.4$, $c\tau=0.1$ mm
 - scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm
 - scA $m_\pi=10$, $m_A=1$, $c\tau=0.1$ mm
 - scA $m_\pi=10$, $m_A=1$, $c\tau=10$ mm
 - scA $m_\pi=10$, $m_A=3.33$, $c\tau=0.1$ mm
 - scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm

$s/\sqrt{b}$ (cumulative, for $|\Delta\eta| < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)

 10^1

- s = Scenario A
 b = MinBias
- scA $m_\pi=4$, $m_A=0.4$, $c\tau=0.1$ mm
 - scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm
 - scA $m_\pi=10$, $m_A=1$, $c\tau=0.1$ mm
 - scA $m_\pi=10$, $m_A=1$, $c\tau=10$ mm
 - scA $m_\pi=10$, $m_A=3.33$, $c\tau=0.1$ mm
 - scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm

0 1 2 3 4 5

Muon SV $|\Delta\eta(\mu, \mu)|$ cut

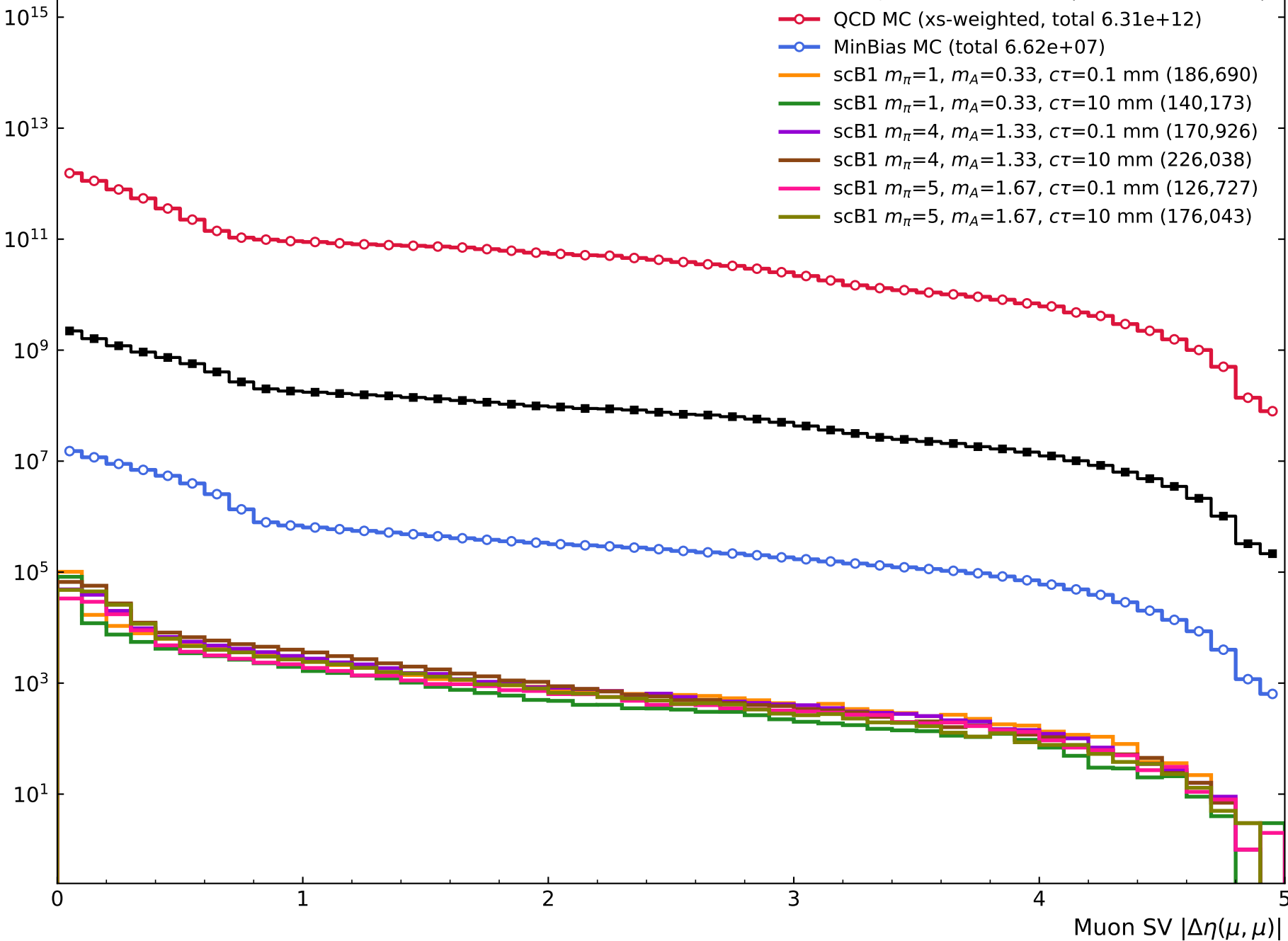
Number of events

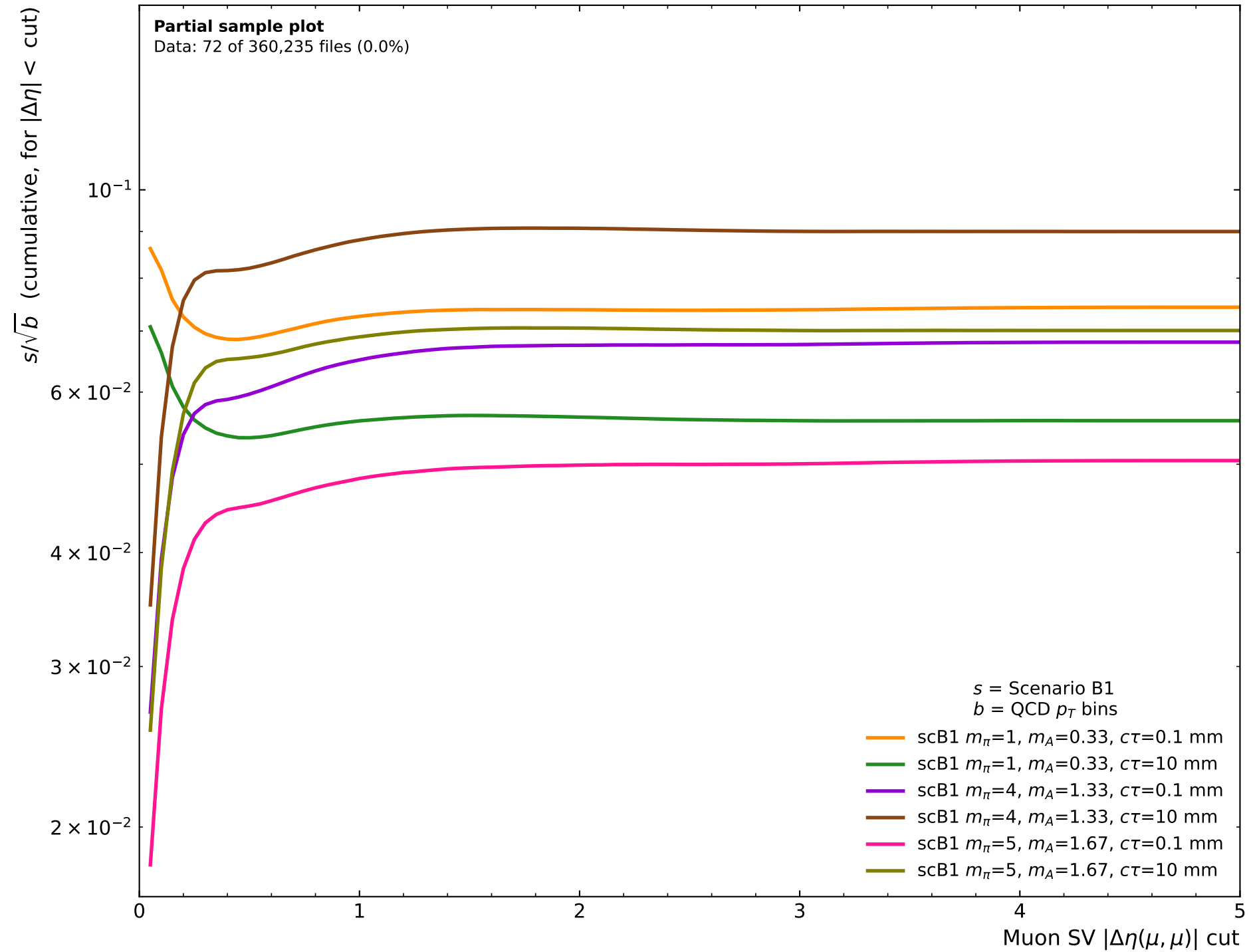
Partial sample plot

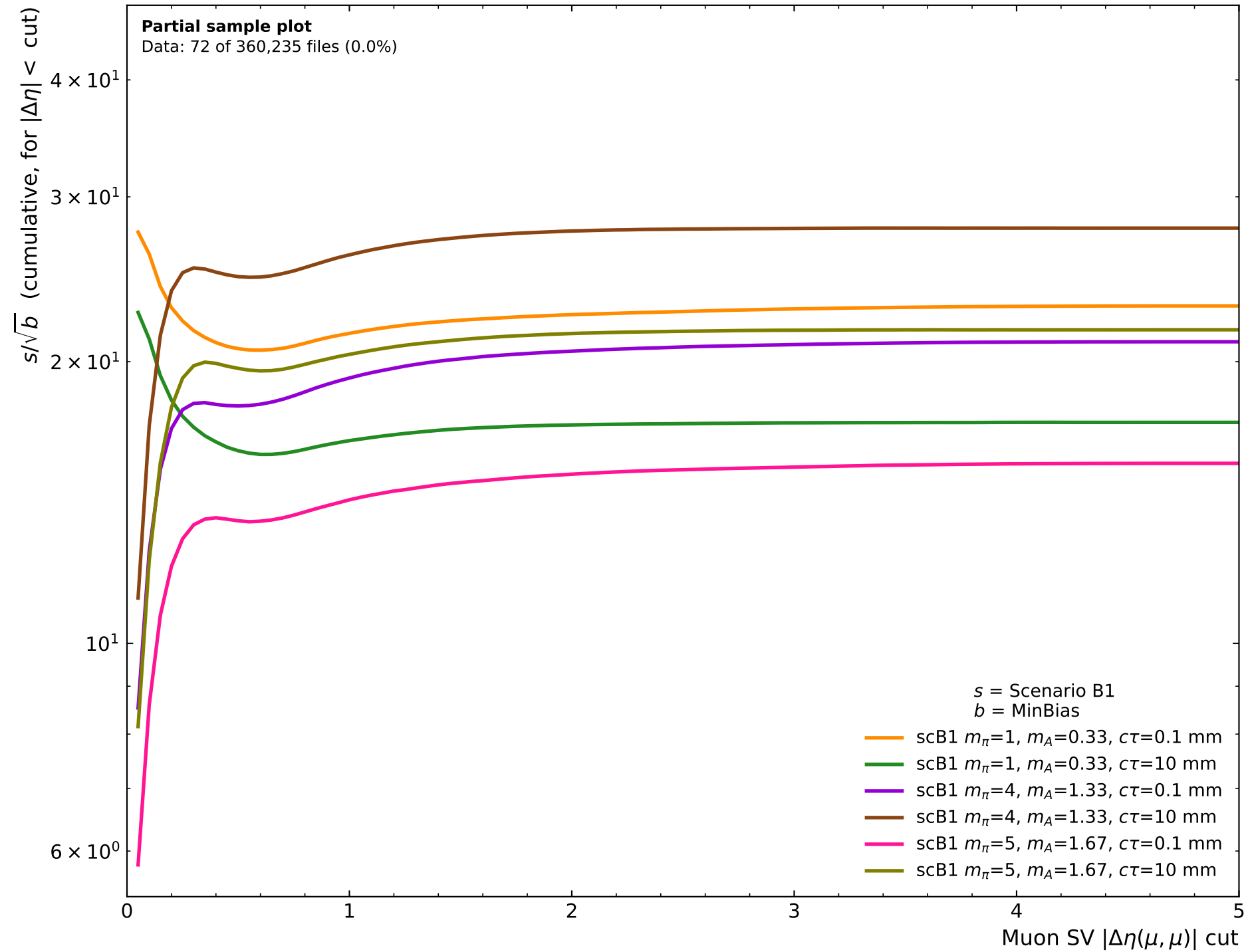
Data: 72 of 360,235 files (0.0%)

Scenario B1

- Data, scaled to 110 fb⁻¹ (est. total 1.07e+10)
- QCD MC (xs-weighted, total 6.31e+12)
- MinBias MC (total 6.62e+07)
- scB1 $m_\pi=1$, $m_A=0.33$, $c\tau=0.1$ mm (186,690)
- scB1 $m_\pi=1$, $m_A=0.33$, $c\tau=10$ mm (140,173)
- scB1 $m_\pi=4$, $m_A=1.33$, $c\tau=0.1$ mm (170,926)
- scB1 $m_\pi=4$, $m_A=1.33$, $c\tau=10$ mm (226,038)
- scB1 $m_\pi=5$, $m_A=1.67$, $c\tau=0.1$ mm (126,727)
- scB1 $m_\pi=5$, $m_A=1.67$, $c\tau=10$ mm (176,043)







Number of events

Partial sample plot

Data: 72 of 360,235 files (0.0%)

- Data, scaled to 110 fb⁻¹ (est. total 1.07e+10)
- QCD MC (xs-weighted, total 6.31e+12)
- MinBias MC (total 6.62e+07)
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm (123,820)
- scB1 $m_\pi=4$, $m_A=1.33$, $c\tau=10$ mm (226,038)

 10^{14} 10^{12} 10^{10} 10^8 10^6 10^4

0.0

0.5

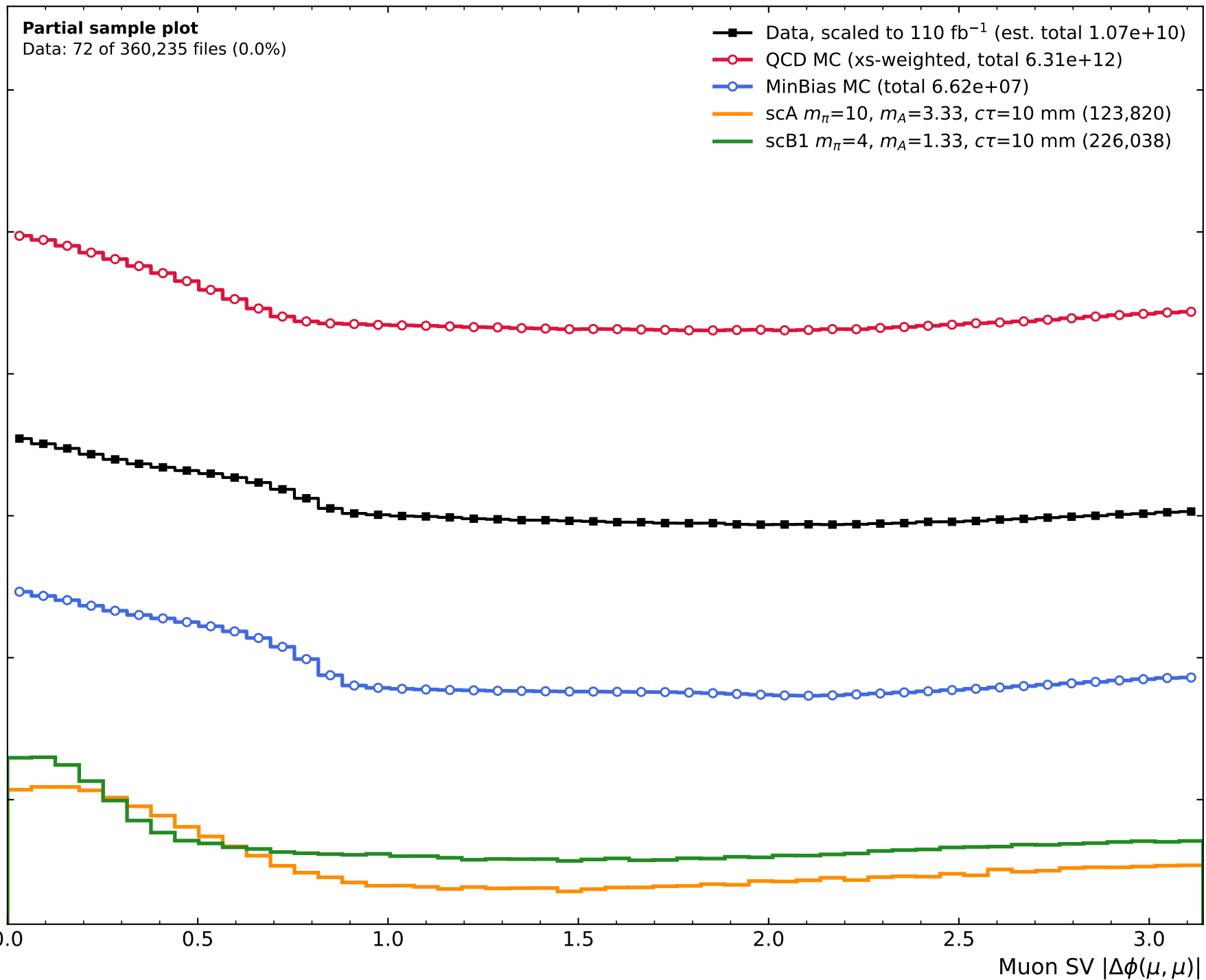
1.0

1.5

2.0

2.5

3.0

Muon SV $|\Delta\phi(\mu, \mu)|$ 

Number of events

Partial sample plot

Data: 72 of 360,235 files (0.0%)

Scenario A

- Data, scaled to 110 fb⁻¹ (est. total 1.07e+10)
- QCD MC (xs-weighted, total 6.31e+12)
- MinBias MC (total 6.62e+07)
- scA $m_\pi=4$, $m_A=0.4$, $c\tau=0.1$ mm (313,115)
- scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm (166,463)
- scA $m_\pi=10$, $m_A=1$, $c\tau=0.1$ mm (184,053)
- scA $m_\pi=10$, $m_A=1$, $c\tau=10$ mm (157,040)
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=0.1$ mm (105,202)
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm (123,820)

10¹⁴

10¹²

10¹⁰

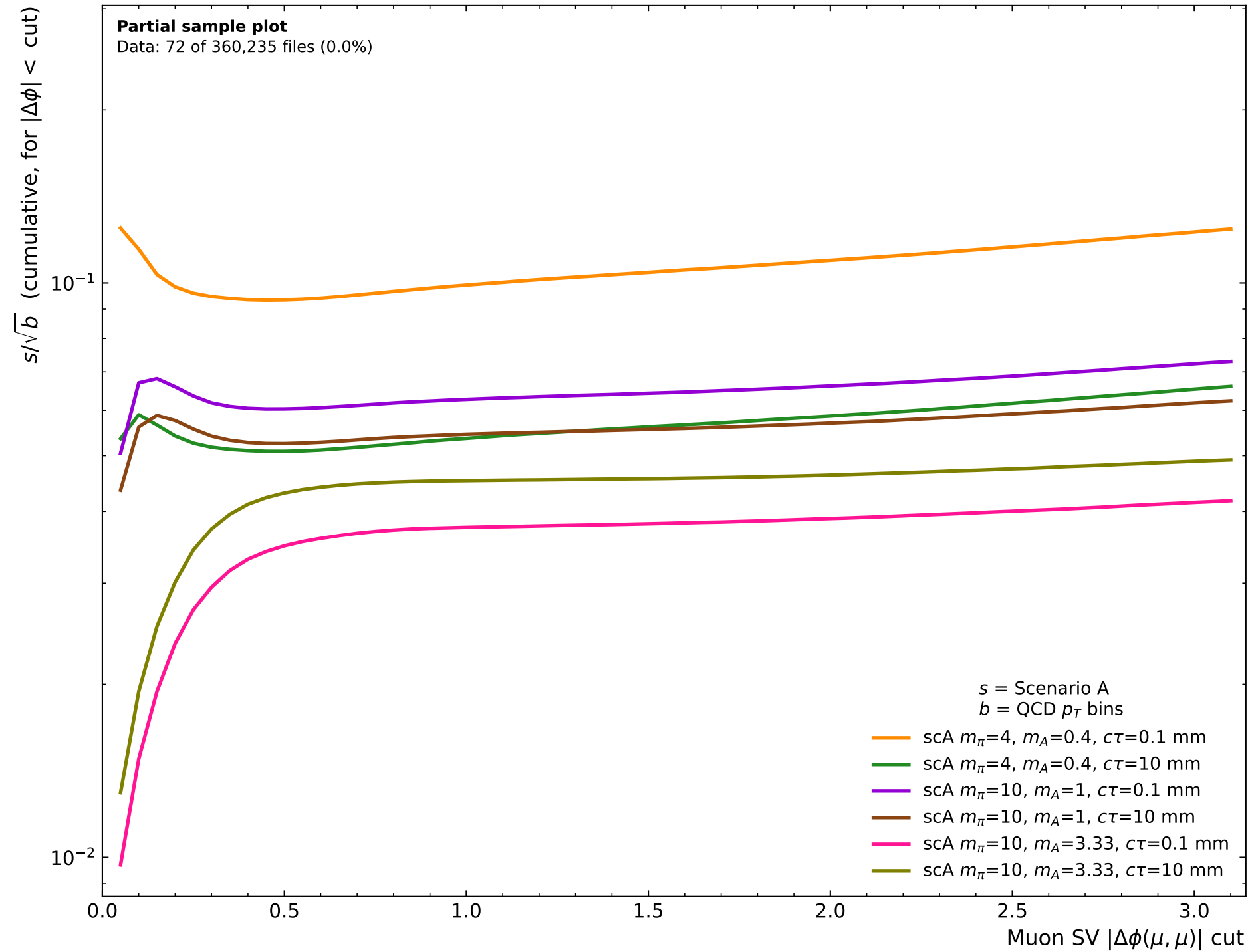
10⁸

10⁶

10⁴

0.0 0.5 1.0 1.5 2.0 2.5 3.0

Muon SV $|\Delta\phi(\mu, \mu)|$



$s/\sqrt{b}$ (cumulative, for $|\Delta\phi| < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)

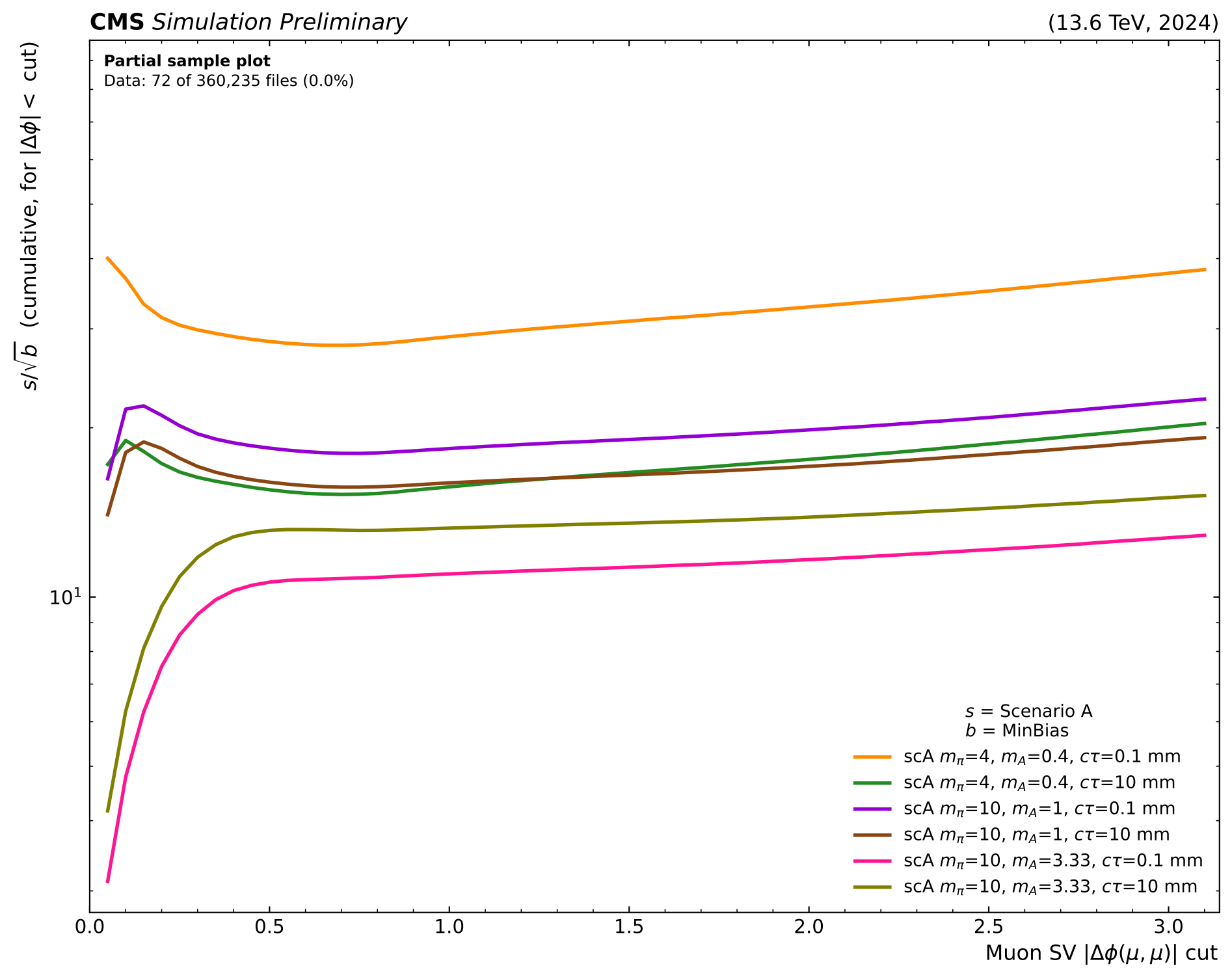
 10^1

0.0 0.5 1.0 1.5 2.0 2.5 3.0

Muon SV $|\Delta\phi(\mu, \mu)|$ cut

s = Scenario A
 b = MinBias

- $\text{scA } m_\pi=4, m_A=0.4, c\tau=0.1 \text{ mm}$
- $\text{scA } m_\pi=4, m_A=0.4, c\tau=10 \text{ mm}$
- $\text{scA } m_\pi=10, m_A=1, c\tau=0.1 \text{ mm}$
- $\text{scA } m_\pi=10, m_A=1, c\tau=10 \text{ mm}$
- $\text{scA } m_\pi=10, m_A=3.33, c\tau=0.1 \text{ mm}$
- $\text{scA } m_\pi=10, m_A=3.33, c\tau=10 \text{ mm}$

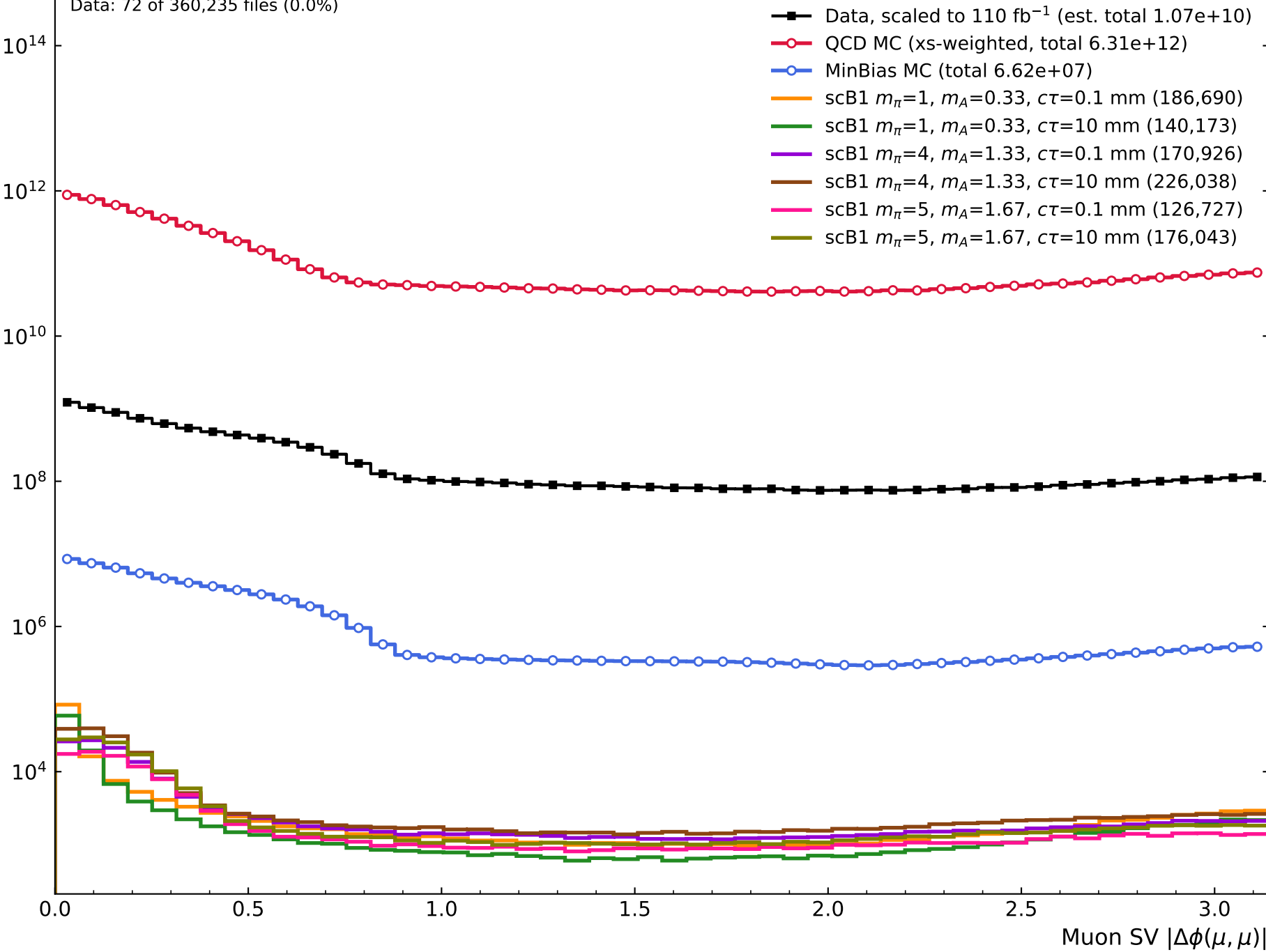


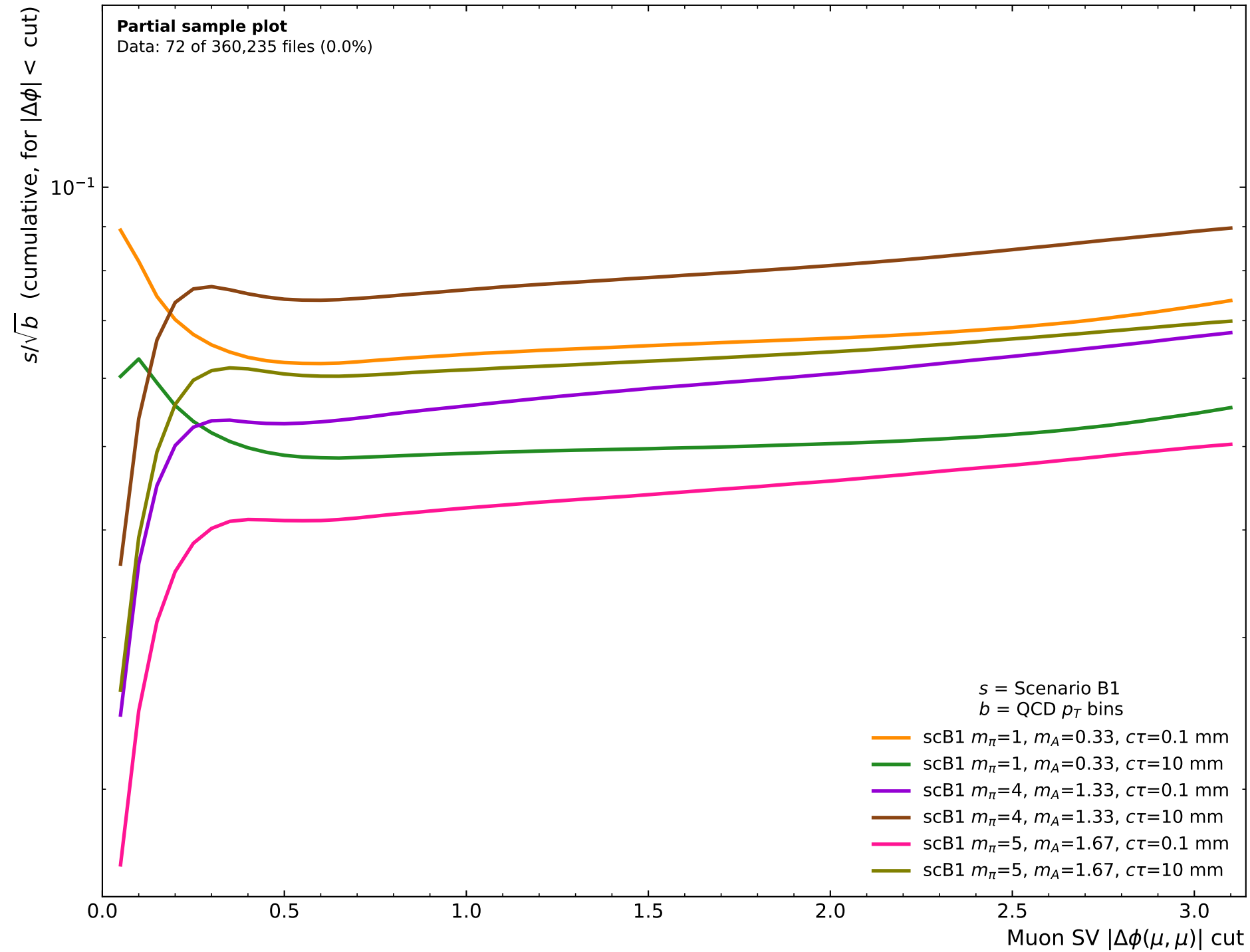
Number of events

Partial sample plot

Data: 72 of 360,235 files (0.0%)

Scenario B1





$s/\sqrt{b}$ (cumulative, for $|\Delta\phi| < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)

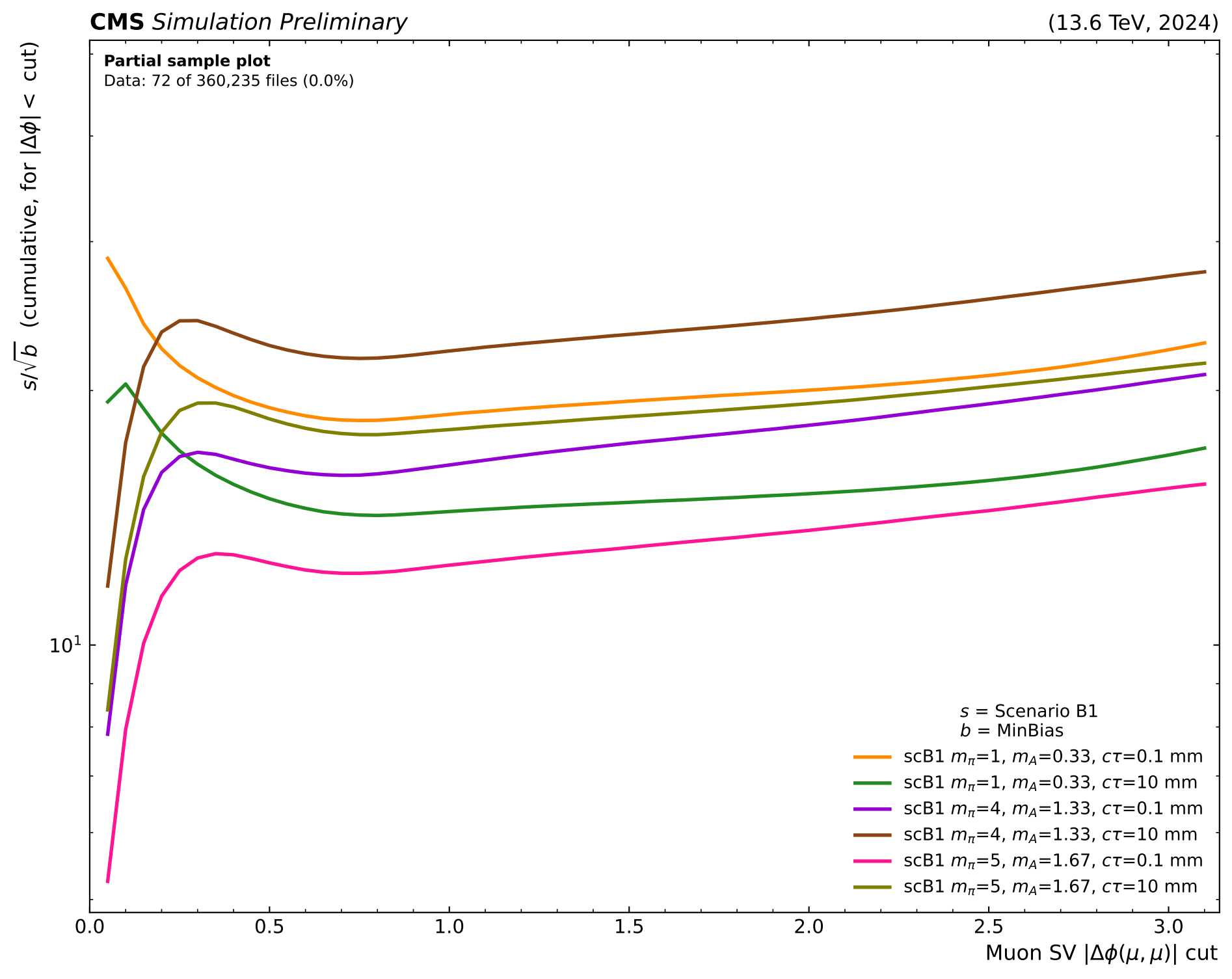
 10^1

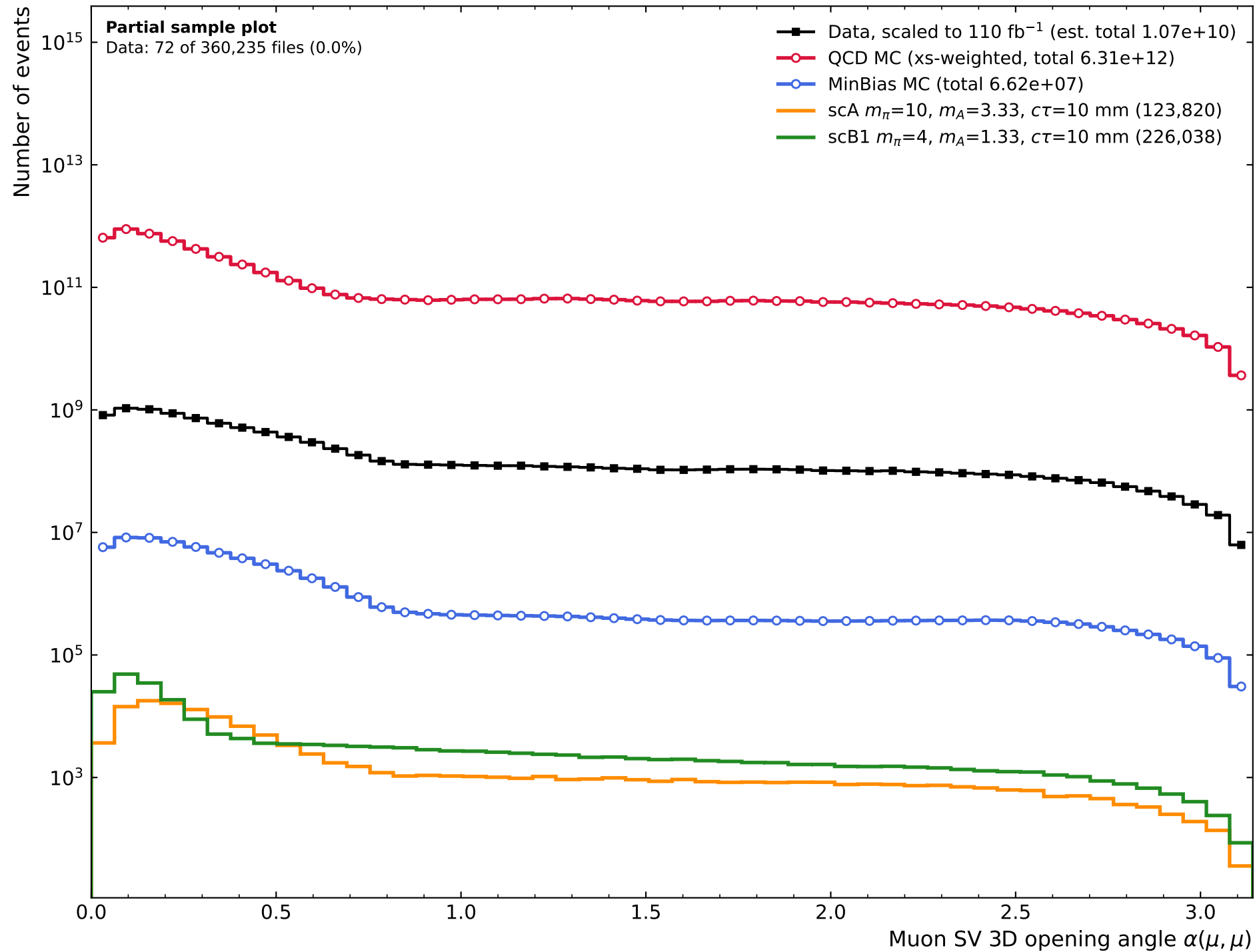
0.0 0.5 1.0 1.5 2.0 2.5 3.0

Muon SV $|\Delta\phi(\mu, \mu)|$ cut

s = Scenario B1
 b = MinBias

- scB1 $m_\pi=1, m_A=0.33, c\tau=0.1$ mm
- scB1 $m_\pi=1, m_A=0.33, c\tau=10$ mm
- scB1 $m_\pi=4, m_A=1.33, c\tau=0.1$ mm
- scB1 $m_\pi=4, m_A=1.33, c\tau=10$ mm
- scB1 $m_\pi=5, m_A=1.67, c\tau=0.1$ mm
- scB1 $m_\pi=5, m_A=1.67, c\tau=10$ mm



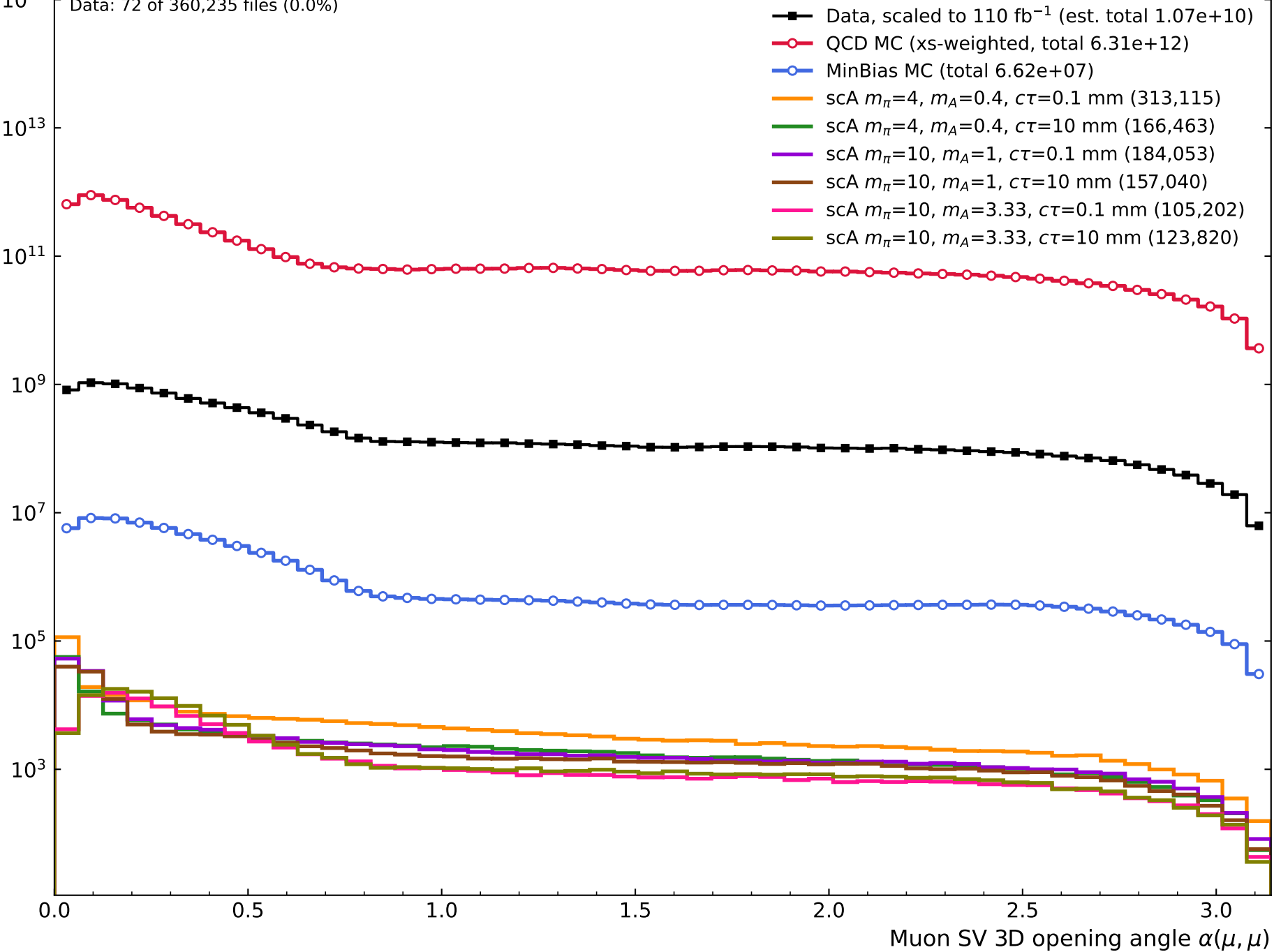


Number of events

Partial sample plot

Data: 72 of 360,235 files (0.0%)

Scenario A



$s/\sqrt{b}$ (cumulative, for $\alpha < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)

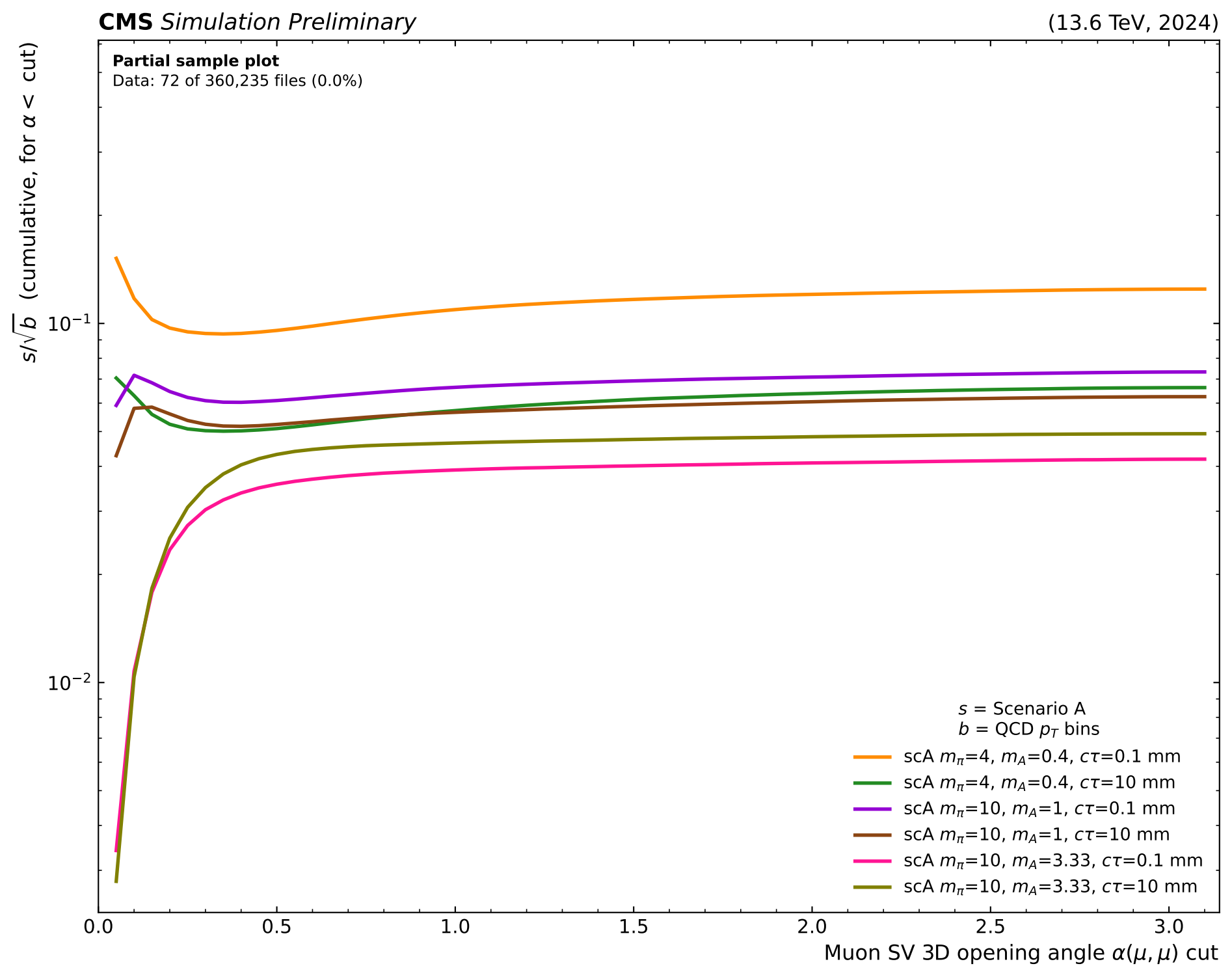
 10^{-2} 10^{-1}

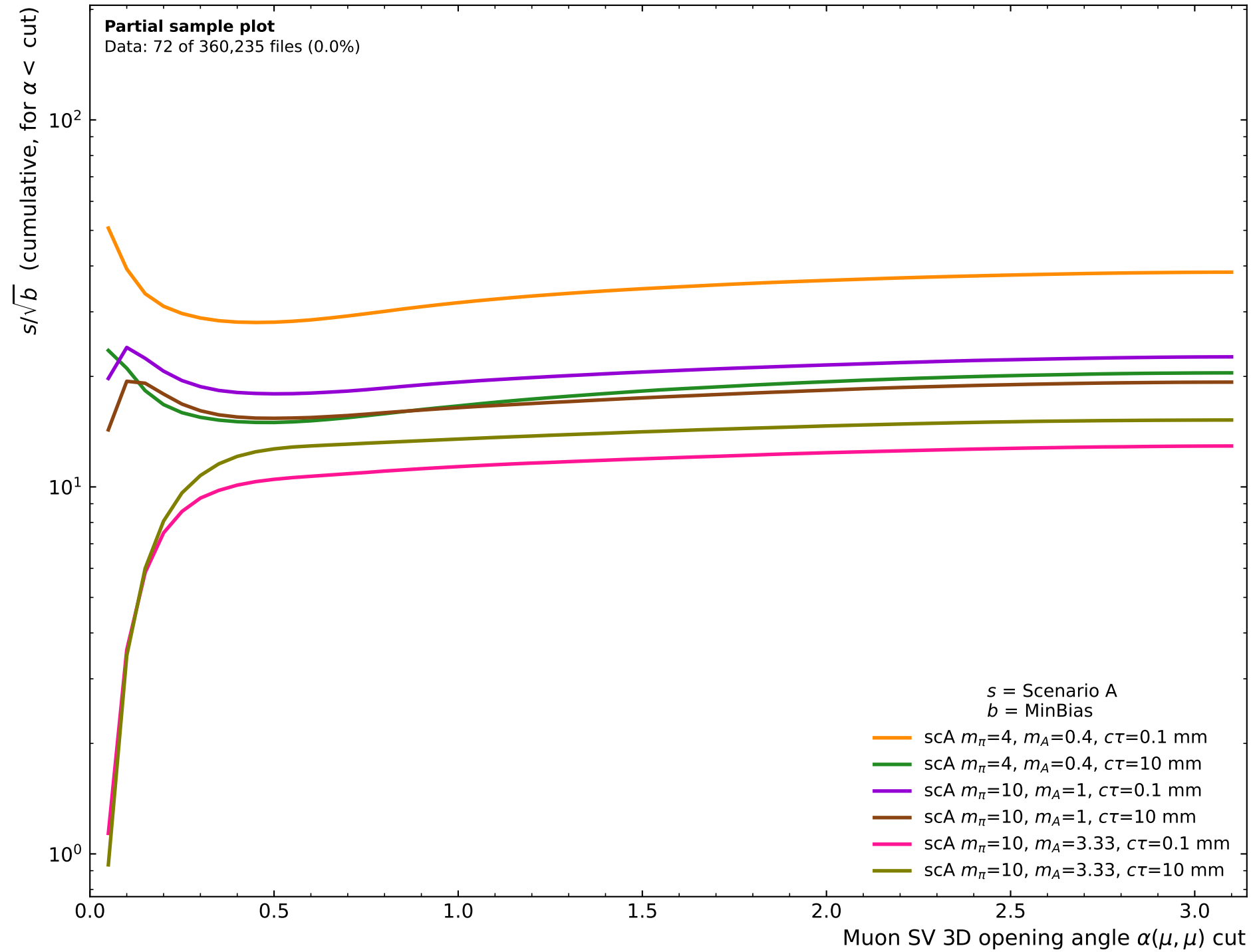
0.0 0.5 1.0 1.5 2.0 2.5 3.0

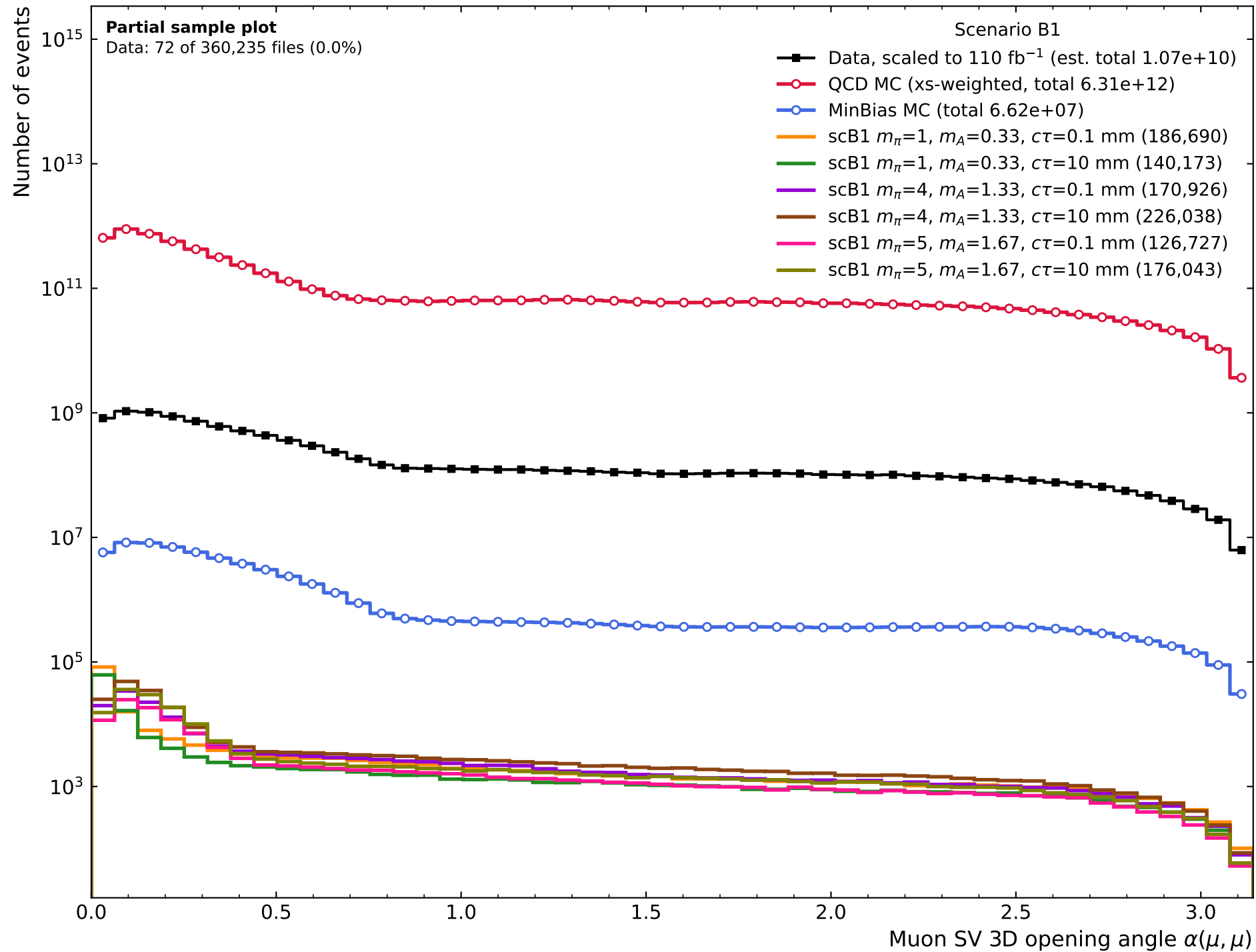
Muon SV 3D opening angle $\alpha(\mu, \mu)$ cut

s = Scenario A
 b = QCD p_T bins

- scA $m_\pi=4$, $m_A=0.4$, $c\tau=0.1$ mm
- scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm
- scA $m_\pi=10$, $m_A=1$, $c\tau=0.1$ mm
- scA $m_\pi=10$, $m_A=1$, $c\tau=10$ mm
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=0.1$ mm
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm



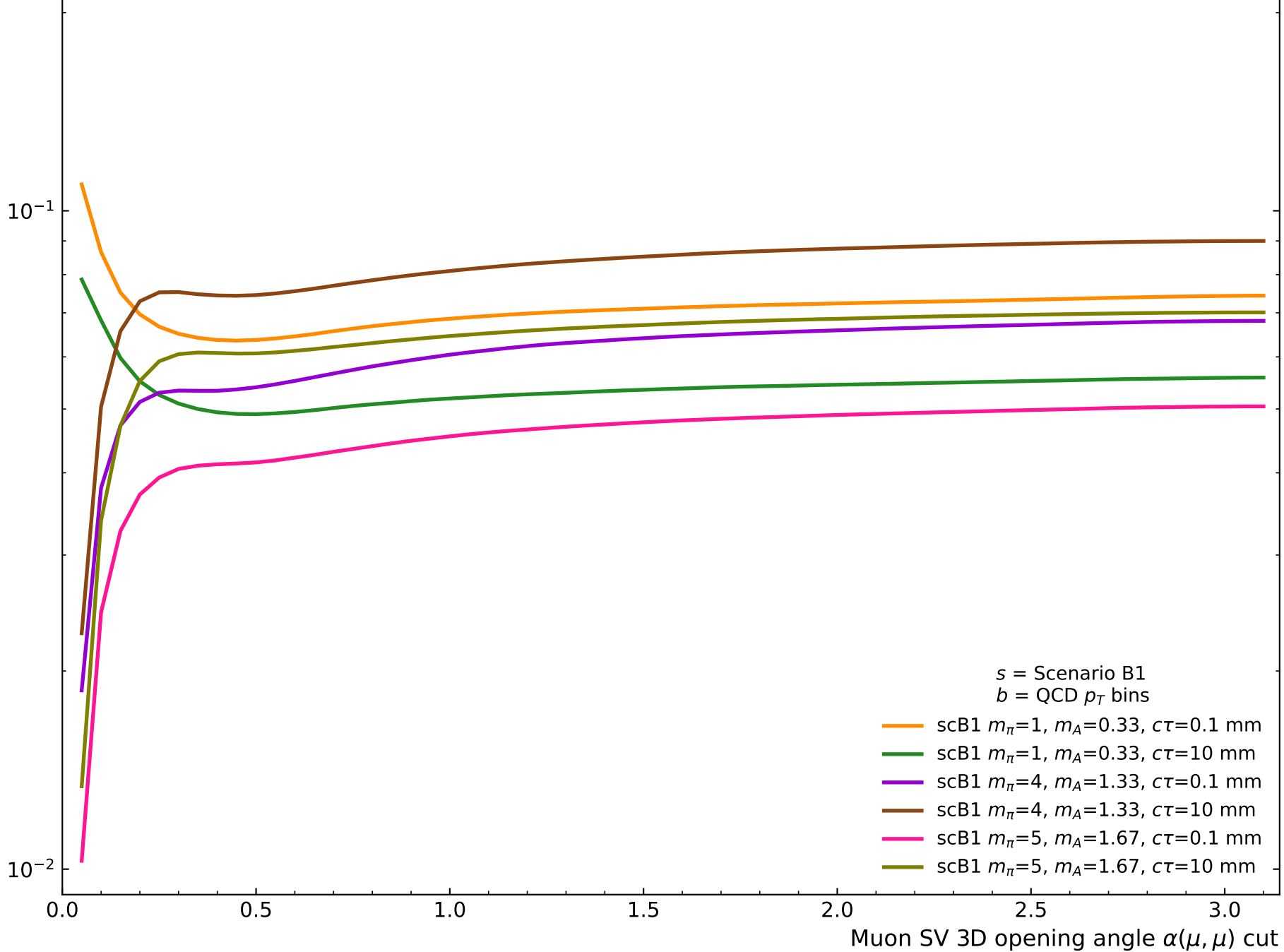




$s/\sqrt{b}$ (cumulative, for $\alpha < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)



$s/\sqrt{b}$ (cumulative, for $\alpha < \text{cut}$)

Partial sample plot

Data: 72 of 360,235 files (0.0%)

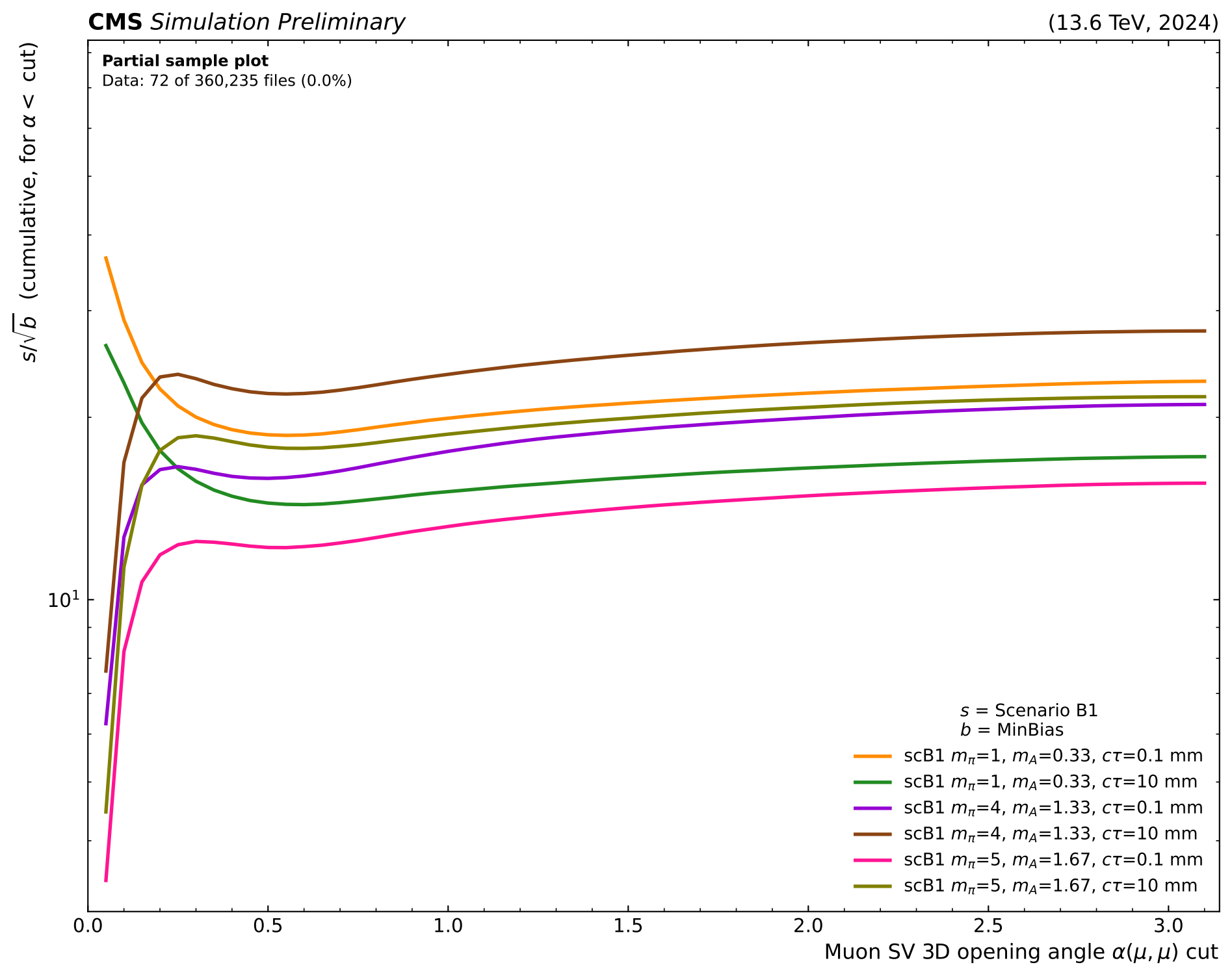
 10^1

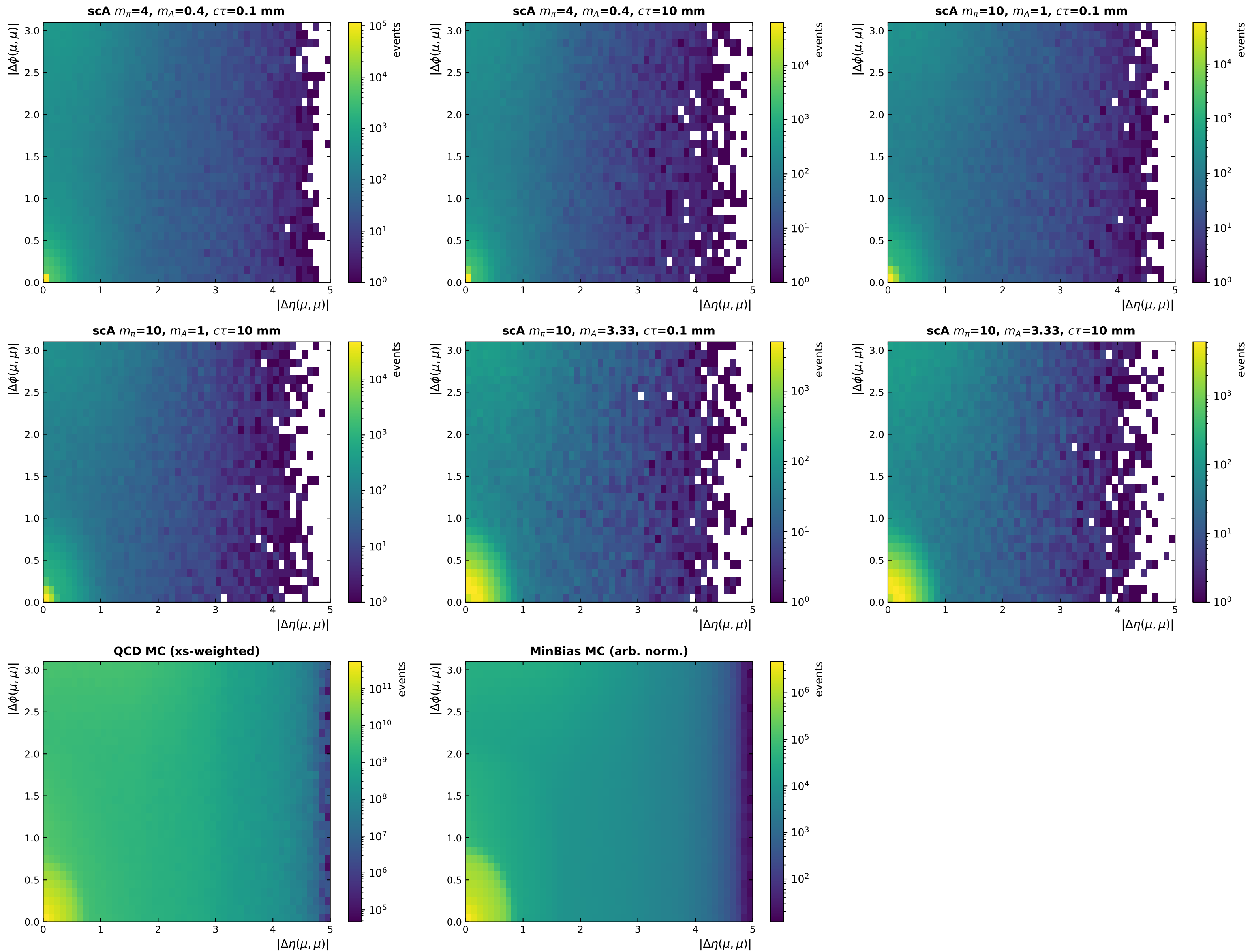
0.0 0.5 1.0 1.5 2.0 2.5 3.0

Muon SV 3D opening angle $\alpha(\mu, \mu)$ cut

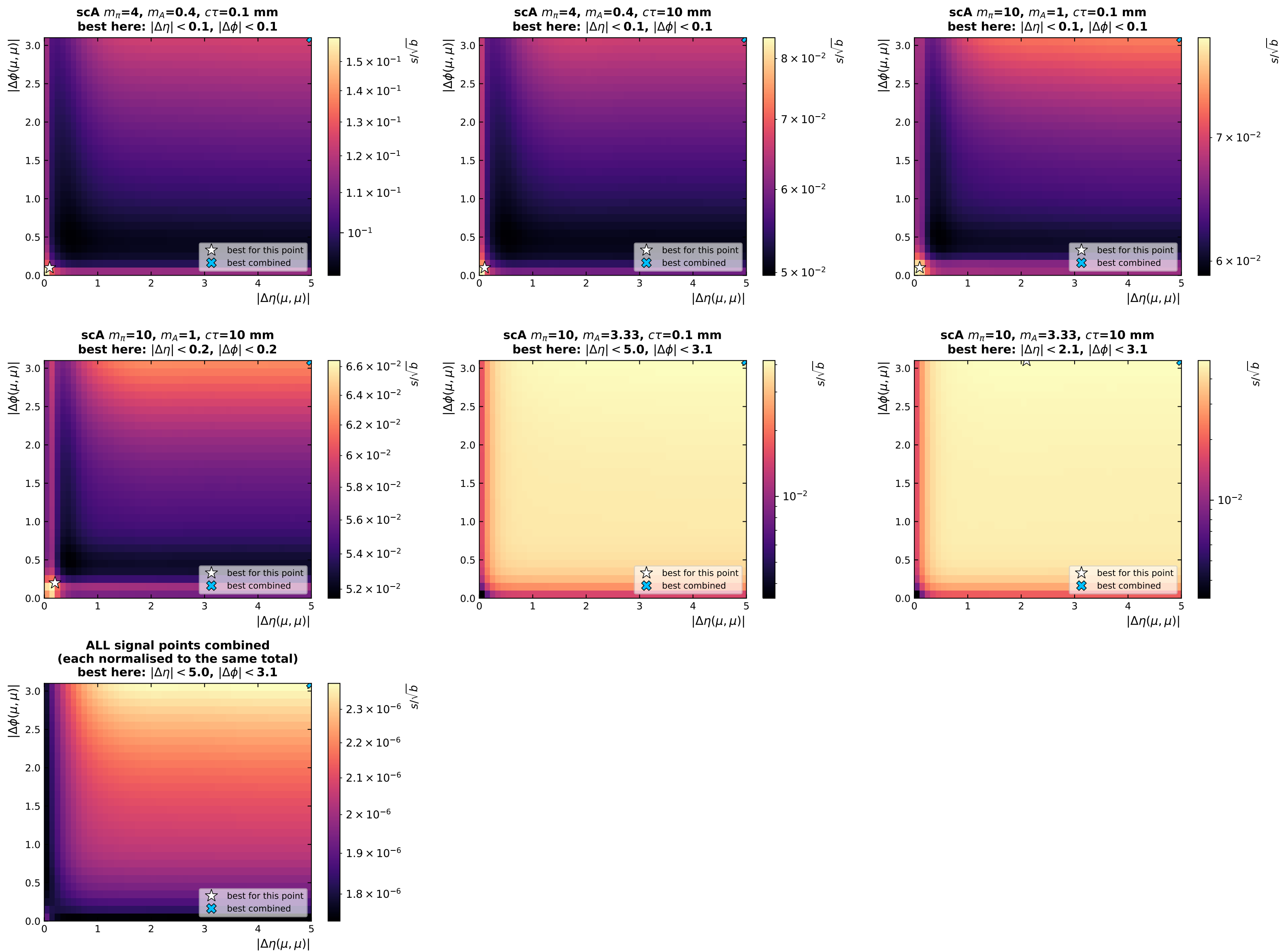
s = Scenario B1
 b = MinBias

- scB1 $m_\pi=1, m_A=0.33, c\tau=0.1$ mm
- scB1 $m_\pi=1, m_A=0.33, c\tau=10$ mm
- scB1 $m_\pi=4, m_A=1.33, c\tau=0.1$ mm
- scB1 $m_\pi=4, m_A=1.33, c\tau=10$ mm
- scB1 $m_\pi=5, m_A=1.67, c\tau=0.1$ mm
- scB1 $m_\pi=5, m_A=1.67, c\tau=10$ mm

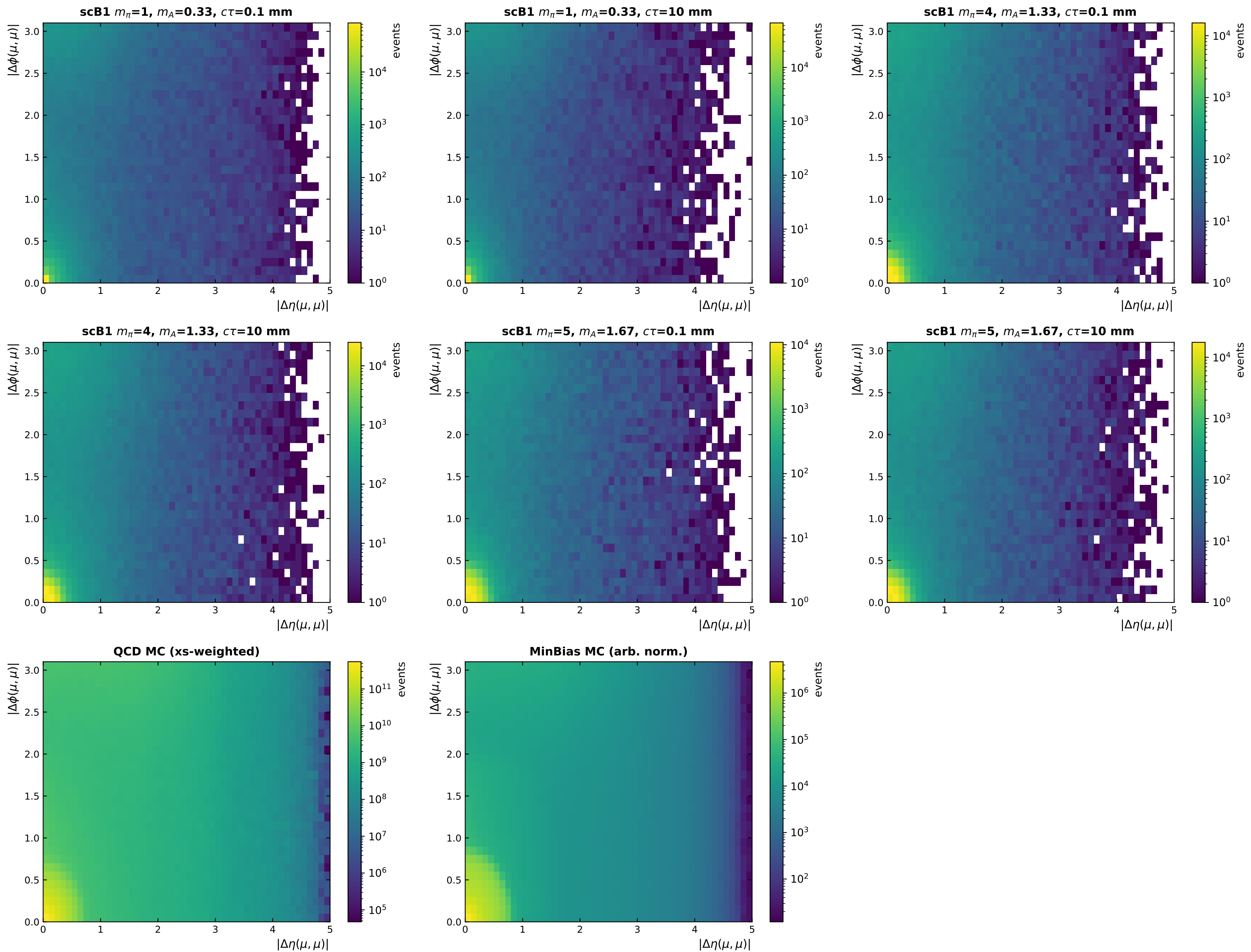


Event density in $(|\Delta\eta|, |\Delta\phi|)$ -- Scenario A

$s/\sqrt{b}$ for a rectangular ($|\Delta\eta|, |\Delta\phi|$) cut -- Scenario A | best combined: $|\Delta\eta| < 5.0, |\Delta\phi| < 3.1$



Event density in $(|\Delta\eta|, |\Delta\phi|)$ -- Scenario B1



$s/\sqrt{b}$ for a rectangular ($|\Delta\eta|, |\Delta\phi|$) cut -- Scenario B1 | best combined: $|\Delta\eta| < 4.8, |\Delta\phi| < 3.1$

